YEAR 12 MATHEMATICS EXTENSION 1

Year 12 Integration Part 3 Further Integration

Lesson and Homework Pack

Revision	Inverse trig, double angles, standard integrals	Y11 Ch 17 · Parts 1–2
Lesson 12A	Inverse trig functions: differentiating	Past papers: 20 marks · 30 min
Lesson 12B	Inverse trig functions: integrating	Past papers: 20 marks · 30 min
Lesson 12C	Further trigonometric integrals	Past papers: 20 marks · 30 min
Lesson 12D	Integration by substitution	Past papers: 20 marks · 30 min
Lesson 12E	Further integration by substitution	Past papers: 18 marks · 25 min
Lesson 12F	Volumes of rotation	Past papers: 21 marks · 30 min
Review	Chapter 12 review exercise	Mixed practice

This is Part 3 of the Year 12 Extension 1 integration course. It follows Chapter 12 of the Year 12 Extension 1 textbook. Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Start with the Revision section: its summary boxes collect the inverse trigonometric functions, the double-angle formulae and the standard integrals from Parts 1 and 2. Answers and full past-paper solutions are in the separate Homework Answers booklet.

REVISION

Inverse trig functions, double angles and standard integrals

A short review of Year 11 Extension 1 Chapter 17 (17A, 17B, 17C and 17E) and of the standard integrals from Parts 1 and 2. Lessons 12A and 12B differentiate and integrate the inverse trigonometric functions, so they need their domains, ranges and graphs. Lesson 12C uses the $\cos 2\theta$ formulae to integrate $\sin^2 x$ and $\cos^2 x$. Lessons 12D to 12F use every standard integral below.

SUMMARY · keep these pages open while you work

INVERSE RELATIONS AND FUNCTIONS (17A)

Suppose that the graph of a relation has been drawn.

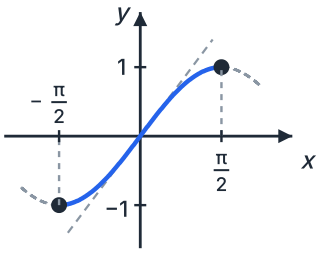
- The graph of the inverse relation is obtained by reflecting the original graph in the diagonal line $y = x$.
- The inverse relation of a given relation is a function if and only if no horizontal line crosses the original graph more than once.
- In particular, suppose that the original relation is a function. If the function is one-to-one, then the inverse relation is also a function. If the function is many-to-one, then the inverse relation is not a function.
- The domain and range of the inverse relation are the range and domain respectively of the original function.
- To find the equation and conditions of the inverse relation, write x for y and y for x every time each variable occurs. If the resulting equation can be solved for y, then the inverse is a function.

Each of the six trigonometric functions fails the horizontal line test, so its inverse relation is not a function. To create an inverse function, we restrict the domain to a piece of the curve between two stationary points (or between two asymptotes for $\tan x$). Because acute angles should be included, the choices below are the obvious ones.

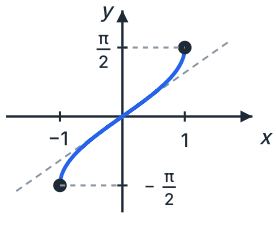
THE DEFINITION OF $y = \sin^{-1} x$ (17B)

- $y = \sin^{-1} x$ is not the inverse relation of $y = \sin x$. It is the inverse function of the restriction of $y = \sin x$ to $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.
- $y = \sin^{-1} x$ has domain $-1 \leq x \leq 1$ and range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
- $y = \sin^{-1} x$ is an increasing function.
- $y = \sin^{-1} x$ has tangent $y = x$ at the origin, and is vertical at its endpoints.

The line $y = x$ is the tangent to $y = \sin x$ at the origin. Reflection in $y = x$ does not move this line, so it is also the tangent to $y = \sin^{-1} x$. The curve $y = \sin x$ is horizontal at its stationary points, so $y = \sin^{-1} x$ is vertical at its endpoints. In the graphs, the dashed diagonal is $y = x$.



$y = \sin x, \ -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

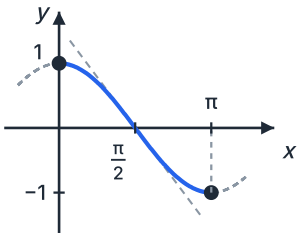


$y = \sin^{-1} x$

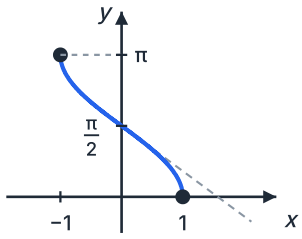
THE DEFINITION OF $y = \cos^{-1} x$ (17B)

- $y = \cos^{-1} x$ is not the inverse relation of $y = \cos x$. It is the inverse function of the restriction of $y = \cos x$ to $0 \leq x \leq \pi$.
- $y = \cos^{-1} x$ has domain $-1 \leq x \leq 1$ and range $0 \leq y \leq \pi$.
- $y = \cos^{-1} x$ is a decreasing function.
- $y = \cos^{-1} x$ has gradient -1 at its y-intercept, and is vertical at its endpoints.

The tangent to $y = \cos x$ at its x-intercept $\left(\frac{\pi}{2}, 0\right)$ is the line $x + y = \frac{\pi}{2}$, with gradient -1. Reflection in $y = x$ maps this line onto itself, so it is also the tangent to $y = \cos^{-1} x$ at its y-intercept $\left(0, \frac{\pi}{2}\right)$. It is dashed in both graphs.



$y = \cos x, \ 0 \leq x \leq \pi$

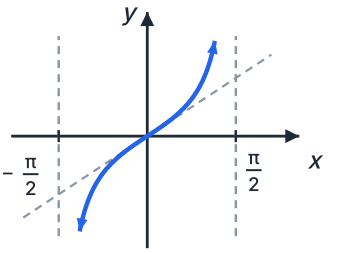


$y = \cos^{-1} x$

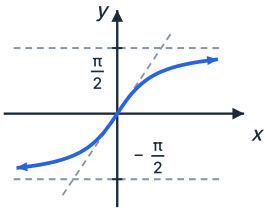
THE DEFINITION OF $y = \tan^{-1} x$ (17B)

- $y = \tan^{-1} x$ is not the inverse relation of $y = \tan x$. It is the inverse function of the restriction of $y = \tan x$ to $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (an open interval).
- $y = \tan^{-1} x$ has domain the real line and range $-\frac{\pi}{2} < y < \frac{\pi}{2}$.
- $y = \tan^{-1} x$ is an increasing function.
- $y = \tan^{-1} x$ has gradient 1 at its y-intercept.
- The lines $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes.

The vertical asymptotes $x = \pm\frac{\pi}{2}$ of $y = \tan x$ reflect onto the horizontal asymptotes $y = \pm\frac{\pi}{2}$ of $y = \tan^{-1} x$. The line $y = x$ (dashed diagonal) is the tangent to both curves at the origin.



$y = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$



$y = \tan^{-1} x$

RADIANS, NOTATION AND INTERPRETING THE RESTRICTIONS (17B)

In this course, radian measure is used exclusively when dealing with the inverse trigonometric functions (except on the calculator, if you leave it in degrees mode). Calculations using degrees should be avoided, or at least not included in the formal working of problems.

- **Use radians** when dealing with inverse trigonometric functions.
- The functions $\operatorname{cosec}^{-1} x$, $\sec^{-1} x$ and $\cot^{-1} x$ are not defined in this course.
- Alternative notation: $\arcsin x = \sin^{-1} x$, $\arccos x = \cos^{-1} x$, $\arctan x = \tan^{-1} x$. Note that $\sin^{-1} x$ is not the reciprocal $(\sin x)^{-1} = \dfrac{1}{\sin x}$.

The key to calculations is to **include the restriction** every time an expression involving the inverse trigonometric functions is rewritten using trigonometric functions:

- $y = \sin^{-1} x$ means $x = \sin y$ where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
- $y = \cos^{-1} x$ means $x = \cos y$ where $0 \leq y \leq \pi$.
- $y = \tan^{-1} x$ means $x = \tan y$ where $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

WORKED EXAMPLE **Textbook Example 4 (17B)**

Find:

a $\cos^{-1}\left(-\frac{1}{2}\right)$

b $\tan^{-1}(-1)$

SOLUTION

a Let $\alpha = \cos^{-1}\left(-\frac{1}{2}\right)$.
Then $\cos\alpha = -\frac{1}{2}$, where $0 \leq \alpha \leq \pi$.
Hence α is in the second quadrant,
and the related angle is $\frac{\pi}{3}$,
so $\alpha = \dfrac{2\pi}{3}$.

b Let $\alpha = \tan^{-1}(-1)$.
Then $\tan\alpha = -1$, where $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$.
Hence α is in the fourth quadrant,
and the related angle is $\frac{\pi}{4}$,
so $\alpha = -\dfrac{\pi}{4}$.

STEPS THE BOOK SKIPS: TEXTBOOK EXAMPLE 4

a $\alpha = \pi - \frac{\pi}{3} = \dfrac{2\pi}{3}$.

WORKED EXAMPLE **Textbook Example 5 (17B)**

Evaluate these expressions exactly, rationalising denominators where necessary.

a $\tan\left(\sin^{-1}\left(-\frac{1}{5}\right)\right)$

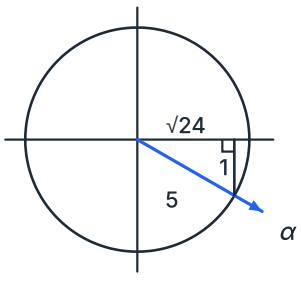
b $\sin\left(\cos^{-1}\frac{4}{5}\right)$

c $\sin\left(\sin^{-1}\frac{4}{5}\right)$

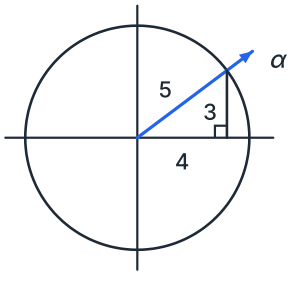
SOLUTION

a Let $\alpha = \sin^{-1}\left(-\frac{1}{5}\right)$.
Then $\sin\alpha = -\frac{1}{5}$, where $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$.
Hence α is in the fourth quadrant,
and

$$\tan\alpha = \frac{-1}{\sqrt{24}} \times \frac{\sqrt{6}}{\sqrt{6}} = -\tfrac{1}{12}\sqrt{6}.$$



b Let $\alpha = \cos^{-1}\frac{4}{5}$.
Then $\cos\alpha = \frac{4}{5}$, where $0 \leq \alpha \leq \pi$.
Hence α is in the first quadrant,
and $\sin\alpha = \frac{3}{5}$.



c Let $\alpha = \sin^{-1}\frac{4}{5}$.
Then $\sin\alpha = \frac{4}{5}$ (where the condition $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$ is now irrelevant).

STEPS THE BOOK SKIPS: TEXTBOOK EXAMPLE 5

a The third side of the triangle is $\sqrt{5^2 - 1^2} = \sqrt{24}$, and $\tan\alpha$ is negative in the fourth quadrant. (Here $\sqrt{24} \times \sqrt{6} = \sqrt{144} = 12$.) So $\tan\alpha = \dfrac{-1}{\sqrt{24}} \times \dfrac{\sqrt{6}}{\sqrt{6}} = \dfrac{-\sqrt{6}}{12}$.

b The third side is $\sqrt{5^2 - 4^2} = 3$.

c So $\sin\left(\sin^{-1}\frac{4}{5}\right) = \frac{4}{5}$.

GRAPHS INVOLVING SHIFTING AND REFLECTING (17C)

Reflecting in either axis, and shifting, can both be applied to the graphs of the three inverse trigonometric functions. As always, substitute key values to confirm the graph. In the case of $\tan^{-1}x$, it is wise to take limits to confirm the horizontal asymptotes.

WORKED EXAMPLE **Textbook Example 6a (17C)**

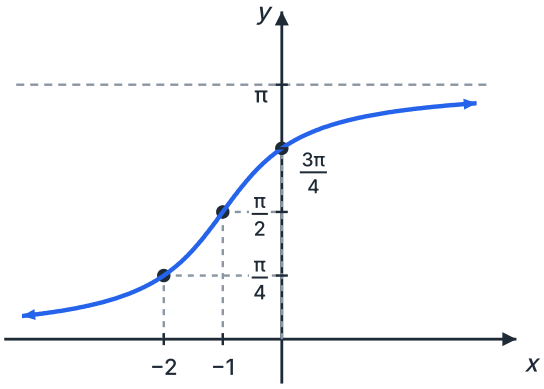
Sketch this graph, stating its domain and range, and any asymptotes:
$y - \frac{\pi}{2} = \tan^{-1}(x+1)$.

SOLUTION

$y - \frac{\pi}{2} = \tan^{-1}(x+1)$ is $y = \tan^{-1}x$ shifted left 1 unit and up $\frac{\pi}{2}$ units.

Confirm this by a small table of values:

x	-2	-1	0
y	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$



The domain is all real numbers, and the range is $0 < y < \pi$.

As $x \to \infty$, $y \to \pi$, and as $x \to -\infty$, $y \to 0$,

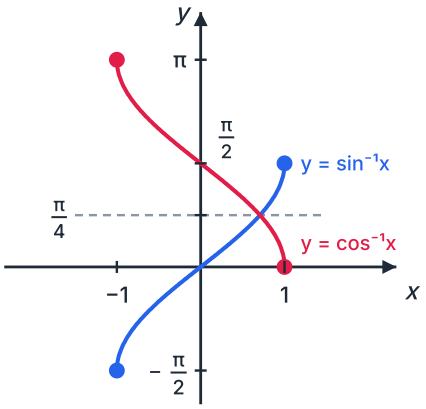
so the horizontal asymptotes are $y = 0$ and $y = \pi$.

STEPS THE BOOK SKIPS: TEXTBOOK EXAMPLE 6A

For example, at $x = -2$: $y = \frac{\pi}{2} + \tan^{-1}(-1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

SYMMETRIES AND THE IDENTITY $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ (17C)

- $y = \sin^{-1} x$ is odd, that is, $\sin^{-1}(-x) = -\sin^{-1} x$.
- $y = \tan^{-1} x$ is odd, that is, $\tan^{-1}(-x) = -\tan^{-1} x$.
- $y = \cos^{-1} x$ has odd symmetry about its y-intercept $\left(0, \frac{\pi}{2}\right)$, that is, $\cos^{-1}(-x) = \pi - \cos^{-1} x$.
- **Complementary angles:** $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, for $-1 \leq x \leq 1$. The graphs are reflections of each other in the horizontal line $y = \frac{\pi}{4}$ (right).



Proof of $\cos^{-1}(-x) = \pi - \cos^{-1} x$.

Let $\alpha = \cos^{-1}(-x)$.
Then $-x = \cos\alpha$,
where $0 \leq \alpha \leq \pi$,
so $\cos(\pi - \alpha) = x$,
because $\cos(\pi - \alpha) = -\cos\alpha$,
$\pi - \alpha = \cos^{-1} x$,
because $0 \leq \pi - \alpha \leq \pi$,
$\alpha = \pi - \cos^{-1} x$.

Proof of $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

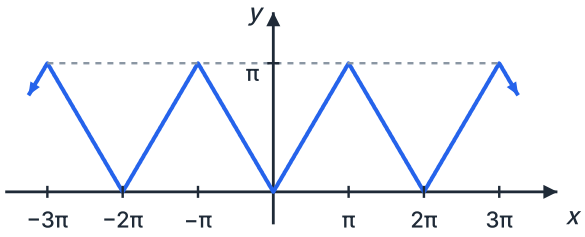
Let $\alpha = \cos^{-1} x$.
Then $x = \cos\alpha$,
where $0 \leq \alpha \leq \pi$,
$\sin\left(\frac{\pi}{2} - \alpha\right) = x$,
because $\sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha$,
$\sin^{-1} x = \frac{\pi}{2} - \alpha$,
because $-\frac{\pi}{2} \leq \frac{\pi}{2} - \alpha \leq \frac{\pi}{2}$,
$\sin^{-1} x + \alpha = \frac{\pi}{2}$
$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

COMPOSITIONS (17C)

- $\sin\left(\sin^{-1}x\right) = x$ and $\cos\left(\cos^{-1}x\right) = x$ for $-1 \le x \le 1$, and $\tan\left(\tan^{-1}x\right) = x$ for all real x.
- The other way round needs the restriction: $\sin^{-1}(\sin x) = x$ only for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$, $\cos^{-1}(\cos x) = x$ only for $0 \le x \le \pi$, and $\tan^{-1}(\tan x) = x$ only for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

The graph of $y = \cos^{-1}(\cos x)$. The domain of this function is the whole real number line, and the graph is far more complicated. A table of values is probably the surest approach, but here is an argument based on symmetries.

A For $0 \le x \le \pi$, $\cos^{-1}(\cos x) = x$, and the graph follows $y = x$.
B Because $\cos x$ is an even function, the graph in the interval $-\pi \le x \le 0$ is the reflection of the graph in the interval $0 \le x \le \pi$.
C We now have the shape of the graph in the interval $-\pi \le x \le \pi$. Because the graph has period 2π, the rest of the graph is just a repetition of this section.



THE DOUBLE-ANGLE FORMULAE (17E)

Replacing both α and β by θ in the compound-angle formula $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$ gives

$$\sin 2\theta = \sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta.$$

The same process gives expansions of $\cos 2\theta$ and $\tan 2\theta$:

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

The $\cos 2\theta$ formulae. Using $\sin^2\theta = 1 - \cos^2\theta$, then $\cos^2\theta = 1 - \sin^2\theta$:

$$\begin{aligned}\cos 2\theta &= \cos^2\theta - \sin^2\theta \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta\end{aligned}$$

These rest on the Pythagorean identities $\sin^2\theta + \cos^2\theta = 1$ and $1 + \tan^2\theta = \sec^2\theta$, which Lesson 12A also uses. Lesson 12C rearranges the last two forms of $\cos 2\theta$ to integrate $\cos^2 x$ and $\sin^2 x$.

WORKED EXAMPLE **Textbook Example 10 (17E)**

Prove that $(\sin x + \cos x)^2 = 1 + \sin 2x$.

SOLUTION

$$\begin{aligned}\text{LHS} &= (\sin x + \cos x)^2 \\ &= \sin^2 x + \cos^2 x + 2\sin x \cos x \\ &= 1 + \sin 2x \\ &= \text{RHS}.\end{aligned}$$

WORKED EXAMPLE **Textbook Example 13 (17E)**

Find $\sin\left(2\cos^{-1}\frac{4}{5}\right)$.

SOLUTION

Let $\alpha = \cos^{-1}\frac{4}{5}$.

Then $\cos\alpha = \frac{4}{5}$, where $0 \le \alpha \le \pi$.

Hence α is in the first quadrant,

$$\begin{aligned}\text{and} \quad \sin 2\alpha &= 2\sin\alpha\cos\alpha \\ &= 2 \times \tfrac{3}{5} \times \tfrac{4}{5} \\ &= \tfrac{24}{25}.\end{aligned}$$

STEPS THE BOOK SKIPS: TEXTBOOK EXAMPLE 13

From the triangle (as in Example 5b), $\sin\alpha = \frac{3}{5}$.

STANDARD INTEGRALS FROM PARTS 1 AND 2

Primitives and the fundamental theorem (4J, 5B, 5E). $F(x)$ is a primitive of $f(x)$ if $F'(x) = f(x)$. If $f(x)$ is continuous in $[a, b]$, then

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \quad \text{where } F(x) \text{ is any primitive of } f(x).$$

Enclose the primitive in square brackets, substitute the upper and lower limits, and subtract.

POWERS (4J, 5E)

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \;\; n \neq -1$$

$$\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C, \;\; n \neq -1$$

REVERSE CHAIN RULE (5I)

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C$$

OR $$\int \big(f(x)\big)^n f'(x)\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C$$

EXPONENTIAL (6D)

$$\int e^x\,dx = e^x + C \qquad \int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$$

RECIPROCAL (6I)

$$\int \frac{1}{x}\,dx = \log_e|x| + C \qquad \int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$$

$$\int \frac{u'}{u}\,dx = \log_e|u| + C \quad \text{OR} \quad \int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)| + C$$

TRIGONOMETRIC (7D)

$$\int \cos x\,dx = \sin x + C \qquad \int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$$

- In every form, C is a constant. Pay attention to the signs in the $\sin$ and $\cos$ integrals.
- For the reverse chain rule, the vital step is to identify u or $f(x)$ and its derivative. For $\dfrac{f'(x)}{f(x)}$, the top must be the derivative of the bottom.
- Radians are the right units for calculus with trigonometric functions.
- **No calculation involving a primitive may cross an asymptote.**

WORKED EXAMPLE **Textbook Example 34 (6I)**

Evaluate these definite integrals using the first two standard forms for reciprocal functions.

a $\displaystyle\int_e^{e^2} \frac{5}{x}\,dx$

b $\displaystyle\int_1^4 \frac{1}{1-2x}\,dx$

c $\displaystyle\int_1^5 \frac{1}{x-2}\,dx$

SOLUTION

a
$$\begin{aligned}\int_e^{e^2} \frac{5}{x}\,dx &= 5\Big[\log_e|x|\Big]_e^{e^2} \\ &= 5\left(\log_e e^2 - \log_e e\right) \\ &= 5(2-1) \\ &= 5\end{aligned}$$

b
$$\begin{aligned}\int_1^4 \frac{1}{1-2x}\,dx &= -\tfrac{1}{2}\Big[\log_e|1-2x|\Big]_1^4 \\ &= -\tfrac{1}{2}\left(\log_e|-7| - \log_e|-1|\right) \\ &= -\tfrac{1}{2}\left(\log_e 7 - 0\right) \\ &= -\tfrac{1}{2}\log_e 7\end{aligned}$$

(here $a = -2$ and $b = 1$)

c This definite integral is meaningless because it crosses the asymptote at $x = 2$.

WORKED EXAMPLE **Textbook Example 17 (7D)**

Evaluate these definite integrals.

a $\int_0^{\pi} \cos x \, dx$

b $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$

c $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sec^2 x \, dx$

SOLUTION

a
$$\begin{aligned}\int_0^{\pi} \cos x \, dx &= \Big[\sin x\Big]_0^{\pi} \\ &= \sin \pi - \sin 0 \\ &= 0\end{aligned}$$

(Use the graph of $y = \sin x$ to see that $\sin \pi = 0$ and $\sin 0 = 0$.)

b
$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sec^2 x \, dx &= \Big[\tan x\Big]_0^{\frac{\pi}{3}} \\ &= \tan \tfrac{\pi}{3} - \tan 0 \\ &= \sqrt{3}\end{aligned}$$

(Here $\tan \frac{\pi}{3} = \sqrt{3}$ and $\tan 0 = 0$.)

c This integral is meaningless because it crosses the asymptote at $x = \frac{\pi}{2}$.

WARM-UP · answers are in the answers booklet

Questions 1 to 4 are from the Year 11 Extension 1 exercise named, with some parts left out. Questions 5 and 6 practise the standard integrals from Parts 1 and 2. Give exact answers, in radians.

1 Exercise 17B Q2

Find the exact value of:

b $\sin^{-1}\frac{1}{2}$

d $\arctan 1$

e $\sin^{-1}(-1)$

f $\cos^{-1}0$

i $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

j $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

k $\arctan\left(-\frac{1}{\sqrt{3}}\right)$

l $\arccos(-1)$

2 Exercise 17B Q5

Find the exact value of:

a $\sin^{-1}\left(\sin\frac{4\pi}{3}\right)$

b $\cos^{-1}\left(\cos\left(-\frac{\pi}{4}\right)\right)$

c $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)$

d $\cos^{-1}\left(\cos\frac{5\pi}{4}\right)$

e $\arcsin\left(2\sin\left(-\frac{\pi}{6}\right)\right)$

f $\arctan\left(3\tan\frac{7\pi}{6}\right)$

3 Exercise 17C Q2

Sketch each function, using appropriate translations of $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$. State the domain and range, and whether it is even, odd or neither.

a $y = \sin^{-1}(x - 1)$

b $y = \cos^{-1}(x + 1)$

c $y - \frac{\pi}{2} = \tan^{-1} x$

4 Exercise 17E Q8

Use a double-angle formula to find the exact value of:

a $\cos\left(2\cos^{-1}\frac{1}{3}\right)$

b $\sin\left(2\cos^{-1}\frac{6}{7}\right)$

c $\tan\left(2\tan^{-1}(-2)\right)$

5 Standard integrals (Parts 1 and 2)

Find each indefinite integral.

a $\int (4x^3 - 6x + 5)\,dx$

b $\int (2x + 1)^4\,dx$

c $\int e^{-2x}\,dx$

d $\int \dfrac{1}{2x + 5}\,dx$

e $\int \sin 4x\,dx$

f $\int \cos \frac{1}{2}x\,dx$

g $\int \sec^2 3x\,dx$

h $\int \dfrac{2x}{x^2 + 1}\,dx$

i $\int 3x^2(x^3 + 1)^4\,dx$

6 Standard integrals (Parts 1 and 2)

Evaluate each definite integral.

a $\int_1^2 (3x^2 - 2x)\,dx$

b $\int_0^1 e^{2x}\,dx$

c $\int_1^e \frac{3}{x}\,dx$

d $\int_0^{\frac{\pi}{6}} \cos 2x\,dx$

e $\int_0^{\frac{\pi}{8}} \sec^2 2x\,dx$

f $\int_0^{\frac{\pi}{3}} \sin x\,dx$

LESSON 12A

Inverse trigonometric functions: differentiating

The inverse trigonometric functions were defined in Chapter 17 of the Year 11 book. We can now apply the normal processes of calculus to them. This section develops their derivatives and applies them to curve-sketching and maximisation. The derivatives come from writing $y = \sin^{-1} x$ as $x = \sin y$, and so on, and the restriction on y decides which square root to take. The chain rule then extends each standard form.

CONCEPT 1 · the derivatives of $\sin^{-1}x$, $\cos^{-1}x$ and $\tan^{-1}x$

DIFFERENTIATING $\sin^{-1} x$ AND $\cos^{-1} x$

To differentiate $y = \sin^{-1} x$ and $y = \cos^{-1} x$, we change to the inverse functions and use the known derivatives of the sine and cosine functions. We need to keep track of the restrictions to the domain so that the choice can be made later between positive and negative square roots.

TWO FACTS FROM YEAR 11

The derivations below use $\cos^2 y + \sin^2 y = 1$, and $\dfrac{dy}{dx} = 1 \Big/ \dfrac{dx}{dy}$.

A Let $y = \sin^{-1} x$.
Then $x = \sin y$, where $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$,
so $\dfrac{dx}{dy} = \cos y$.

Because y is in the first or fourth quadrant, $\cos y$ is positive, so

$$\cos y = +\sqrt{1 - \sin^2 y}$$
$$= \sqrt{1 - x^2} \quad (\text{since } \sin y = x).$$

Thus $\dfrac{dx}{dy} = \sqrt{1 - x^2}$, so

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}.$$

Hence $\dfrac{d}{dx}\sin^{-1} x = \dfrac{1}{\sqrt{1 - x^2}}$.

B Let $y = \cos^{-1} x$.
Then $x = \cos y$, where $0 \le y \le \pi$,
so $\dfrac{dx}{dy} = -\sin y$.

Because y is in the first or second quadrant, $\sin y$ is positive, so

$$\sin y = +\sqrt{1 - \cos^2 y}$$
$$= \sqrt{1 - x^2} \quad (\text{since } \cos y = x).$$

Thus $\dfrac{dx}{dy} = -\sqrt{1 - x^2}$, so

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}}.$$

Hence $\dfrac{d}{dx}\cos^{-1} x = -\dfrac{1}{\sqrt{1 - x^2}}$.

DIFFERENTIATING $\tan^{-1} x$

The problem of which square root to choose does not arise when differentiating $y = \tan^{-1} x$.

Let $y = \tan^{-1} x$.
Then $x = \tan y$, where $-\frac{\pi}{2} < y < \frac{\pi}{2}$,
so

$$\frac{dx}{dy} = \sec^2 y$$
$$= 1 + \tan^2 y.$$

Hence $\frac{dx}{dy} = 1 + x^2$ (since $\tan y = x$),
and $\frac{dy}{dx} = \frac{1}{1+x^2}$,
giving the standard form

$$\frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}.$$

1 STANDARD FORMS FOR DIFFERENTIATION

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}$$

THE ENDPOINTS OF $\sin^{-1} x$ AND $\cos^{-1} x$

The formulae for $\sin^{-1} x$ and $\cos^{-1} x$ hold only for $-1 < x < 1$. At $x = \pm 1$ the denominator $\sqrt{1-x^2}$ is zero, so the derivative is undefined. The graphs are vertical at their endpoints, as the Revision section noted. The formula for $\tan^{-1} x$ holds for every real x.

TUTOR DEMO **Product rule and a linear inside function**

Differentiate these functions.
a $y = x^2 \cos^{-1} x$
b $y = \cos^{-1}(ax + b)$

YOUR TURN **Same skill**

Differentiate these functions.

a $y = x^2 \sin^{-1} x$

b $y = \tan^{-1}(ax + b)$

WORKED EXAMPLE 1 **Keep for homework**

Differentiate these functions.

a $y = x \tan^{-1} x$

b $y = \sin^{-1}(ax + b)$

SOLUTION

a $y = x \tan^{-1} x$

Let $u = x$ and $v = \tan^{-1} x$.

Then $u' = 1$ and $v' = \dfrac{1}{1 + x^2}$.

$$y' = vu' + uv'$$

$$= \tan^{-1} x \times 1 + x \times \frac{1}{1 + x^2}$$

$$= \tan^{-1} x + \frac{x}{1 + x^2}$$

b $y = \sin^{-1}(ax + b)$

Let $u = ax + b$,

then $y = \sin^{-1} u$.

Hence $\dfrac{du}{dx} = a$

and $\dfrac{dy}{du} = \dfrac{1}{\sqrt{1 - u^2}}$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1 - (ax + b)^2}} \times a$$

$$= \frac{a}{\sqrt{1 - (ax + b)^2}}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 1

In the last line, u is replaced by $ax + b$.

CONCEPT 2 · linear extensions, and tangents with a given gradient

LINEAR EXTENSIONS

The method used in part **b** above can be applied to all three inverse trigonometric functions, giving a further set of standard forms.

THE OTHER TWO, BY THE SAME METHOD

With $u = ax + b$ and $\dfrac{du}{dx} = a$, the chain rule gives

$$\frac{d}{dx}\cos^{-1}(ax+b) = -\frac{1}{\sqrt{1-u^2}} \times a = -\frac{a}{\sqrt{1-(ax+b)^2}},$$

$$\frac{d}{dx}\tan^{-1}(ax+b) = \frac{1}{1+u^2} \times a = \frac{a}{1+(ax+b)^2}.$$

2 FURTHER STANDARD FORMS FOR DIFFERENTIATION

$$\frac{d}{dx}\sin^{-1}(ax+b) = \frac{a}{\sqrt{1-(ax+b)^2}}$$

$$\frac{d}{dx}\cos^{-1}(ax+b) = -\frac{a}{\sqrt{1-(ax+b)^2}}$$

$$\frac{d}{dx}\tan^{-1}(ax+b) = \frac{a}{1+(ax+b)^2}$$

FINDING WHERE THE TANGENT HAS A GIVEN GRADIENT

Put y' equal to the given gradient and solve for x. The square root is cleared by squaring, so check each solution in the domain. Then find each y-value from the equation of the curve.

TUTOR DEMO **Tangents of gradient 2**

a Find the points A and B on the curve $y = \sin^{-1}(x+1)$ where the tangent has gradient 2.

b Sketch the curve, showing these points.

YOUR TURN **Same method**

a Find the points A and B on the curve $y = \cos^{-1}(x-2)$ where the tangent has gradient -2.

b Sketch the curve, showing these points.

WORKED EXAMPLE 2 **Keep for homework**

a Find the points A and B on the curve $y = \cos^{-1}(x-1)$ where the tangent has gradient -2.

b Sketch the curve, showing these points.

SOLUTION

a Differentiating, $y' = -\dfrac{1}{\sqrt{1-(x-1)^2}}$.

Put $y' = -2$. Then

$$-\frac{1}{\sqrt{1-(x-1)^2}} = -2$$

$$1-(x-1)^2 = \tfrac{1}{4}$$

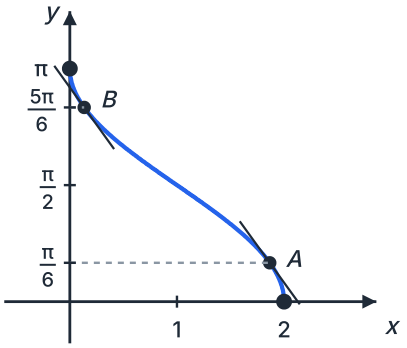
$$(x-1)^2 = \tfrac{3}{4}$$

$$x-1 = \tfrac{1}{2}\sqrt{3} \text{ or } -\tfrac{1}{2}\sqrt{3}$$

$$x = 1+\tfrac{1}{2}\sqrt{3} \text{ or } 1-\tfrac{1}{2}\sqrt{3},$$

so the points are $A\left(1+\frac{1}{2}\sqrt{3}, \frac{\pi}{6}\right)$ and $B\left(1-\frac{1}{2}\sqrt{3}, \frac{5\pi}{6}\right)$.

b



STEPS THE BOOK SKIPS: WORKED EXAMPLE 2

a From $\dfrac{1}{\sqrt{1-(x-1)^2}} = 2$,

$$\sqrt{1-(x-1)^2} = \tfrac{1}{2}.$$

When $x = 1+\frac{1}{2}\sqrt{3}$, $y = \cos^{-1}\frac{1}{2}\sqrt{3} = \frac{\pi}{6}$.

When $x = 1-\frac{1}{2}\sqrt{3}$, $y = \cos^{-1}\left(-\frac{1}{2}\sqrt{3}\right) = \frac{5\pi}{6}$.

b The domain is $-1 \le x-1 \le 1$, that is, $0 \le x \le 2$, and the range is $0 \le y \le \pi$. The endpoints are $(0, \pi)$ and $(2, 0)$. The short lines at A and B are the tangents of gradient -2.

CONCEPT 3 · functions whose derivatives are zero are constants

PROVING IDENTITIES BY DIFFERENTIATION

Several identities involving inverse trigonometric functions can be obtained by showing that some derivative is zero, and hence that the original function is a constant. The following identity is the clearest example. It has been proven already in Section 17C of the Year 11 book using symmetry arguments.

THE METHOD

- Show that the derivative of the function is zero for every x in its domain.
- Then the function is a constant C on that domain.
- Find C by substituting any convenient value of x in the domain.

TUTOR DEMO **An odd function**

a Differentiate $\sin^{-1} x + \sin^{-1}(-x)$.
b Hence prove the identity $\sin^{-1}(-x) = -\sin^{-1} x$.

YOUR TURN **Same method**

a Differentiate $\tan^{-1} x + \tan^{-1}(-x)$.
b Hence prove the identity $\tan^{-1}(-x) = -\tan^{-1} x$.

WORKED EXAMPLE 3 **Keep for homework**

a Differentiate $\sin^{-1} x + \cos^{-1} x$.
b Hence prove the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

SOLUTION

a

$$\begin{aligned}\frac{d}{dx}\left(\sin^{-1} x + \cos^{-1} x\right) &= \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} \\ &= 0\end{aligned}$$

b Hence $\sin^{-1} x + \cos^{-1} x = C$, for some constant C.
Substitute $x = 0$, then $0 + \frac{\pi}{2} = C$,
so $C = \frac{\pi}{2}$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, as required.

CONCEPT 4 · curve sketching using calculus

CURVE SKETCHING USING CALCULUS

The usual methods of curve sketching can now be extended to curves whose equations involve the inverse trigonometric functions. The next worked example applies calculus to sketching the curve $y = \cos^{-1} \cos x$, which was sketched without calculus in Section 17C of the Year 11 book.

THE METHOD

- A simple table of test values gives some key points.
- The derivative gives the shape of the curve joining these points.
- Remember that $\sqrt{A^2} = |A|$. So $\sqrt{\sin^2 x} = \sin x$ when $\sin x$ is positive, and $\sqrt{\sin^2 x} = -\sin x$ when $\sin x$ is negative.

TUTOR DEMO **A sawtooth**

Use calculus to sketch $y = \tan^{-1} \tan x$.

YOUR TURN **Same method**

Use calculus to sketch $y = \cos^{-1} \cos 2x$.

WORKED EXAMPLE 4 **Keep for homework**

Use calculus to sketch $y = \cos^{-1}\cos x$.

SOLUTION

The function is periodic with the same period 2π as $\cos x$. A simple table of test values gives some key points. The shape of the curve joining these points can be obtained by calculus.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$	3π	$\ldots$
y	0	$\frac{\pi}{2}$	π	$\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π	$\ldots$

Differentiating using the chain rule,

$$\frac{dy}{dx} = \frac{\sin x}{\sqrt{1-\cos^2 x}}$$
$$= \frac{\sin x}{\sqrt{\sin^2 x}}.$$

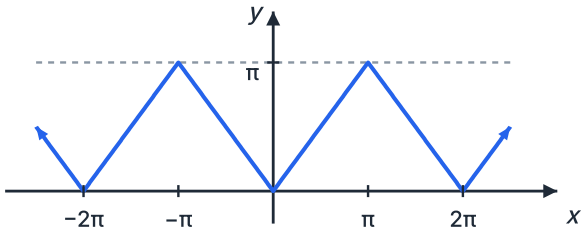
Let $u = \cos x$,
then $y = \cos^{-1} u$.
Hence $\frac{du}{dx} = -\sin x$
and $\frac{dy}{du} = -\frac{1}{\sqrt{1-u^2}}$.

When $\sin x$ is positive, $\sqrt{\sin^2 x} = \sin x$, so $\frac{dy}{dx} = 1$.

When $\sin x$ is negative, $\sqrt{\sin^2 x} = -\sin x$, so $\frac{dy}{dx} = -1$.

$$\text{Hence } \frac{dy}{dx} = \begin{cases} 1, & \text{for } x \text{ in quadrants 1 and 2,} \\ -1, & \text{for } x \text{ in quadrants 3 and 4.} \end{cases}$$

This means that the graph consists of a series of intervals, each with gradient 1 or -1.



STEPS THE BOOK SKIPS: WORKED EXAMPLE 4

The first line of the chain rule, in full:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{1}{\sqrt{1-\cos^2 x}} \times (-\sin x) = \frac{\sin x}{\sqrt{1-\cos^2 x}}.$$

CONCEPT 5 · chain-rule extensions to the standard forms

CHAIN-RULE EXTENSIONS

The usual chain-rule extensions to the standard forms can be used with the inverse trigonometric forms. They provide an alternative to the chain-rule setting out.

WHERE THE FORMS COME FROM

For example, if $y = \sin^{-1} u$, where u is a function of x, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{\sqrt{1-u^2}} \times u' = \frac{u'}{\sqrt{1-u^2}}.$$

3 CHAIN-RULE EXTENSIONS TO THE STANDARD FORMS

$$\frac{d}{dx}\sin^{-1} u = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}\cos^{-1} u = -\frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}\tan^{-1} u = \frac{u'}{1+u^2}$$

OR

$$\frac{d}{dx}\sin^{-1} f(x) = \frac{f'(x)}{\sqrt{1-\left(f(x)\right)^2}}$$

$$\frac{d}{dx}\cos^{-1} f(x) = -\frac{f'(x)}{\sqrt{1-\left(f(x)\right)^2}}$$

$$\frac{d}{dx}\tan^{-1} f(x) = \frac{f'(x)}{1+\left(f(x)\right)^2}$$

TUTOR DEMO **Using Box 3**

Use the chain-rule extension formulae to differentiate:

a $\tan^{-1} e^{-2x}$

b $\cos^{-1}(x^2-1)$

YOUR TURN **Same method**

Use the chain-rule extension formulae to differentiate:
a $\tan^{-1} e^{-4x}$
b $\sin^{-1}(1-x^2)$

WORKED EXAMPLE 5 **Keep for homework**

Use the chain-rule extension formulae to differentiate $\tan^{-1} e^{-5x}$.

SOLUTION

Here $u = e^{-5x}$ and $u' = -5e^{-5x}$,

so $$\frac{d}{dx}\left(\tan^{-1} e^{-5x}\right) = \frac{u'}{1+u^2} = \frac{-5e^{-5x}}{1+e^{-10x}}.$$

OR here $f(x) = e^{-5x}$ and $f'(x) = -5e^{-5x}$,

so $$\frac{d}{dx}\left(\tan^{-1} e^{-5x}\right) = \frac{f'(x)}{1+\left(f(x)\right)^2} = \frac{-5e^{-5x}}{1+e^{-10x}}.$$

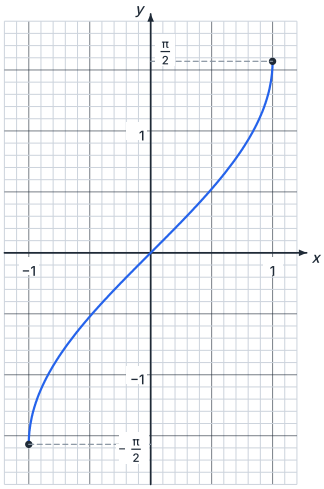
STEPS THE BOOK SKIPS: WORKED EXAMPLE 5

In the last line, $u^2 = \left(e^{-5x}\right)^2 = e^{-10x}$.

TEXTBOOK HOMEWORK · Exercise 12A

FOUNDATION

1 a Photocopy the graph of $y = \sin^{-1} x$ shown to the right. Then carefully draw a tangent at each x value in the table. Then, by measurement and calculation of rise/run, find the gradient of each tangent correct to two decimal places and fill in the second row of the table.



x	-1	-0.7	-0.5	-0.2	0	0.3	0.6	0.8	1
$\frac{dy}{dx}$									

b Check your gradients using $\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}}$.

2 Differentiate with respect to x:

a $\cos^{-1} x$

b $\tan^{-1} x$

c $\sin^{-1} 2x$

d $\tan^{-1} 3x$

e $\cos^{-1} 5x$

f $\sin^{-1}(-x)$

g $\sin^{-1} x^2$

h $\tan^{-1} x^3$

i $\tan^{-1}(x+2)$

j $\cos^{-1}(1-x)$

k $x \sin^{-1} x$

l $(1+x^2)\tan^{-1} x$

m $\sin^{-1} \frac{1}{5}x$

n $\tan^{-1} \frac{1}{4}x$

o $\cos^{-1} \sqrt{x}$

p $\tan^{-1} \sqrt{x}$

q $\tan^{-1} \frac{1}{x}$

3 Find the gradient of the tangent to each curve at the point indicated:

a $y = 2\tan^{-1} x$, at $x = 0$

b $y = \sqrt{3}\sin^{-1} x$, at $x = \frac{1}{2}$

c $y = \tan^{-1} 2x$, at $x = -\frac{1}{2}$

d $y = \cos^{-1} \frac{x}{2}$, at $x = \sqrt{3}$

4 Find, in the form $y = mx + b$, the equation of the tangent and the normal to each curve at the point indicated:

a $y = 2\cos^{-1} 3x$, at $x = 0$

b $y = \sin^{-1} \frac{x}{2}$, at $x = \sqrt{2}$

5 a Show that $\frac{d}{dx}\left(\sin^{-1} x + \cos^{-1} x\right) = 0$.

b Hence explain why $\sin^{-1} x + \cos^{-1} x$ is a constant function, and use any convenient value of x in its domain to find the value of the constant.

6 Use the method of the previous question to show that each of these functions is a constant function, and find the value of the constant.

a $\cos^{-1} x + \cos^{-1}(-x)$

b $2\sin^{-1}\sqrt{x} - \sin^{-1}(2x-1)$

DEVELOPMENT

7 a If $f(x) = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2)$, show that $f''(x) = \dfrac{1}{1+x^2}$.

b Is the graph of $y = f(x)$ concave up or concave down at $x = -1$?

8 Show that the gradient of the curve $y = \dfrac{\sin^{-1} x}{x}$ at the point where $x = \frac{1}{2}$ is $\frac{2}{3}\left(2\sqrt{3} - \pi\right)$.

9 Find the derivative of each function in simplest form:

a $x\cos^{-1}x-\sqrt{1-x^2}$

b $\sin^{-1}e^{3x}$

c $\sin^{-1}\frac{1}{4}(2x-3)$

d $\tan^{-1}\dfrac{1}{1-x}$

e $\sin^{-1}e^{x}$

f $\log_e\sqrt{\sin^{-1}x}$

g $\sin^{-1}\sqrt{\log_e x}$

h $\sqrt{x}\,\sin^{-1}\sqrt{1-x}$

i $\tan^{-1}\dfrac{x+2}{1-2x}$

10

a (i) If $y=\left(\sin^{-1}x\right)^2$, show that $y''=\dfrac{2+\dfrac{2x\sin^{-1}x}{\sqrt{1-x^2}}}{1-x^2}$.

(ii) Hence show that $(1-x^2)y''-xy'-2=0$.

b Show that $y=e^{\sin^{-1}x}$ satisfies the differential equation $(1-x^2)y''-xy'-y=0$.

11 Consider the function $y = \sin^{-1} 2x$.

a Write down the range of the function.

b Make x the subject of the equation.

c Find $\dfrac{dx}{dy}$, and explain why it is never negative.

d Use the result that $\dfrac{dy}{dx}$ is the reciprocal of $\dfrac{dx}{dy}$ to find $\dfrac{dy}{dx}$.

12 Use the approach in the previous question to find $\dfrac{dy}{dx}$ given:

a $y = \sin^{-1} \frac{x}{2}$ **b** $y = \cos^{-1}(x - 1)$ **c** $y = \tan^{-1} \sqrt{x}$

13 Consider the function $f(x) = \cos^{-1} x^2$.

a What is the domain of $f(x)$?

b About which line is the graph of $y = f(x)$ symmetric?

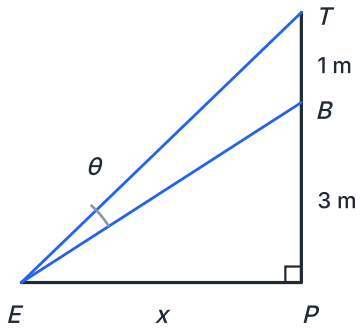
c Find $f'(x)$.

d Show that $y = f(x)$ has a maximum turning point at $x = 0$.

e Show that $f'(x)$ is undefined at the endpoints of the domain. What is the geometrical significance of this?

f Sketch the graph of $y = f(x)$.

14 A picture TB that is 1 metre tall is hung on a wall so that its bottom edge B is 3 metres above the eye E of a viewer. Let the distance EP be x metres, and let θ be the angle that the picture subtends at E.

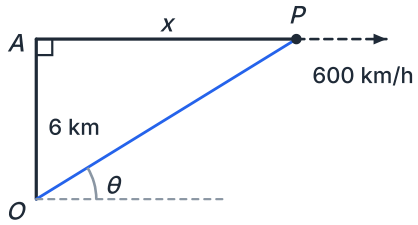


a Show that $\theta = \tan^{-1} \dfrac{4}{x} - \tan^{-1} \dfrac{3}{x}$.

b Show that θ is maximised when the viewer is $2\sqrt{3}$ metres from the wall.

c Show that the maximum angle subtended by the picture at E is $\tan^{-1} \dfrac{\sqrt{3}}{12}$.

15 A plane P at a constant altitude of 6 km and at a constant speed of 600 km/h is flying directly away from an observer at O on the ground. The point A on the path of the plane lies directly above O. Let the distance AP be x km, and let the angle of elevation of the plane from the observer be θ.



a Show that $\theta = \tan^{-1}\dfrac{6}{x}$.

b Show that $\dfrac{d\theta}{dt} = \dfrac{-3600}{x^2+36}$ radians per hour.

c Hence find, in radians per second, the rate at which θ is decreasing at the instant when the distance AP is 3 km.

16

a State the domain of $f(x) = \tan^{-1} x + \tan^{-1}\dfrac{1}{x}$, and its symmetry.

b Show that $f'(x) = 0$ for all values of x in the domain.

17 Consider the function $f(x) = \cos^{-1}\dfrac{1}{x}$.

a State the domain of $f(x)$. (Hint: Think about it rather than relying on algebra.)

b Recalling that $\sqrt{x^2} = |x|$, show that $f'(x) = \dfrac{1}{|x|\sqrt{x^2-1}}$.

c Comment on $f'(1)$ and $f'(-1)$.

d Use the expression for $f'(x)$ in part **b** to write down separate expressions for $f'(x)$ when $x > 1$ and when $x < -1$.

e Explain why $f(x)$ is increasing for $x > 1$ and for $x < -1$.

f Find:

(i) $\lim\limits_{x\to\infty} f(x)$ (ii) $\lim\limits_{x\to-\infty} f(x)$

g Sketch the graph of $y = f(x)$.

18 Use the formulae for the chain-rule standard forms in Box 3 to differentiate:

a $\tan^{-1} e^{3x}$ b $\sin^{-1} x^3$ c $\cos^{-1}(\log_e x)$

ENRICHMENT

19 a What is the domain of $g(x) = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$?

b Show that $g'(x) = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{x}{|x|\sqrt{1-x^2}}$.

c Hence determine the interval over which $g(x)$ is constant, and find this constant.

20 In question 9 i, you proved that $\dfrac{d}{dx}\tan^{-1}\dfrac{x+2}{1-2x}$ was $\dfrac{1}{1+x^2}$, which is also the derivative of $\tan^{-1} x$. What is going on?

21 The function $f(x)$ is defined by the rule $f(x) = \sin^{-1}(\sin x)$.

a State the domain and range of $f(x)$, and whether it is even, odd or neither.

b Show that $f'(x) = \dfrac{\cos x}{|\cos x|}$.

c Is $f'(x)$ defined when $\cos x = 0$?

d What are the only two values that $f'(x)$ takes if $\cos x \neq 0$, and when does each of these values occur?

e Sketch the graph of $f(x)$ using the above information and a table of values if necessary.

22 In Question 14, construct the circle that passes through T and B and is tangent to the horizontal line ℓ through P. Then the point E on ℓ at which TB subtends the greatest angle is the point where the circle touches the line ℓ.

a Prove this using Euclidean geometry.

b Explain how to construct this circle, and E, using straight edge and compasses.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q4** **1 mark**

What is the derivative of $\sin^{-1} 3x$?

A. $-\dfrac{1}{\sqrt{1-9x^2}}$

B. $\dfrac{3}{\sqrt{1-9x^2}}$

C. $-\dfrac{1}{\sqrt{1-3x^2}}$

D. $-\dfrac{3}{\sqrt{1-9x^2}}$

Hint: which standard form does this look like?

PP 2 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q9** **1 mark**

What is the derivative of $x\cos^{-1}(x) - \sqrt{1-x^2}$?

A. $\dfrac{-1}{\sqrt{1-x^2}}$

B. $\dfrac{-2}{\sqrt{1-x^2}}$

C. $\cos^{-1} x$

D. $\sin^{-1} x$

Hint: what is the first thing you need to differentiate?

PP 3 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(a)** **2 marks**

Find $\dfrac{d}{dx}\left(\cos^{-1}(3x^2)\right)$.

Hint: which standard form does this look like?

PP 4 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(g)** 2 marks

Find $\dfrac{dy}{dx}$ if $y = e^{-x}\sin^{-1}x$

Hint: what is the first thing you need to differentiate?

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(d)** 2 marks

Show that the normal to the curve $y = \sin^{-1}\left(x + \dfrac{\pi}{2}\right) + \dfrac{\pi}{2}$ at the point where $x = -\dfrac{\pi}{2}$ passes through the origin.

Hint: what are the coordinates of the point on the curve?

PP 6 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(a)(i)(ii)** 3 marks

(i) Show that $\dfrac{d}{dx}\left(x\tan^{-1}x\right) = \tan^{-1}x + \dfrac{x}{1+x^2}$. 1

(ii) Hence, find the exact value of $\displaystyle\int_0^{\sqrt{3}} \tan^{-1}x\,dx$. 2

Hint: what does part (i) give you?

PP 7 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q11(d)** 3 marks

Show that $\dfrac{d}{dx}\left(\dfrac{\sin^{-1}(3x)}{3x}\right) = \dfrac{3x - \sin^{-1}(3x)\sqrt{1-9x^2}}{3x^2\sqrt{1-9x^2}}$

Hint: what is the first thing you need to differentiate?

PP 8 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)** 3 marks

The gradient of the tangent to the curve $g(x) = \left(\sin^{-1}\dfrac{x}{b}\right)^2$ at the point A with x coordinate $\dfrac{b}{2}$ is $\dfrac{2\pi}{3}$.
Find the value of b.

Hint: what does the given gradient tell you?

PP 9 **Ascham School, 2020, Trial HSC Examination, Mathematics Extension 1, Q14(c)** 3 marks

Given that $f(x) = \cos^{-1}(2x-1) - 2\cos^{-1}\sqrt{x}$ for $0 \leq x \leq 1$, show that $f'(x) = 0$.

Hint: what is the derivative of each term?

LESSON 12B

Inverse trigonometric functions: integrating

This lesson deals with the integrals associated with the inverse trigonometric functions, and with the standard applications of those integrals. Reversing the derivatives of Lesson 12A gives standard forms for $\dfrac{1}{\sqrt{a^2-x^2}}$ and $\dfrac{1}{a^2+x^2}$. When x^2 has a coefficient, take it out first. The reverse chain rule extends the forms, and a product-rule derivative gives the primitive of $\sin^{-1}x$ itself.

CONCEPT 1 · the basic standard forms

REVERSING THE DERIVATIVES

Differentiation of the inverse trigonometric functions yields purely algebraic functions,

$$\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}} \qquad \text{and} \qquad \frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2}.$$

These are remarkable results, and indicate once again that trigonometric functions are very closely related to algebraic functions associated with squares and square roots. The relationship was already clear when the trigonometric functions were defined using the circle, whose equation $x^2+y^2=r^2$ is Pythagoras' theorem, which is purely algebraic.

This lesson concerns integration, and we begin by reversing the three standard forms for differentiation.

THE THIRD STANDARD FORM

Lesson 12A also gave $\dfrac{d}{dx}\cos^{-1}x=\dfrac{-1}{\sqrt{1-x^2}}$, so $\dfrac{d}{dx}\left(-\cos^{-1}x\right)=\dfrac{1}{\sqrt{1-x^2}}$ as well.

4 STANDARD FORMS FOR INTEGRATION

$$\int\frac{1}{\sqrt{1-x^2}}\,dx=\sin^{-1}x+C \qquad \text{OR} \qquad \int\frac{1}{\sqrt{1-x^2}}\,dx=-\cos^{-1}x+C$$

$$\int\frac{1}{1+x^2}\,dx=\tan^{-1}x+C$$

TWO ANSWERS THAT AGREE

The two primitives of $\dfrac{1}{\sqrt{1-x^2}}$ differ only by a constant. From the identity $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$ (revision page), $-\cos^{-1}x=\sin^{-1}x-\frac{\pi}{2}$. So the constant $-\frac{\pi}{2}$ is absorbed into C, and a definite integral gives the same value with either form (see the example below).

ALGEBRAIC INTEGRANDS

Thus some purely algebraic functions require the inverse trigonometric functions for their integration. We have seen this sort of phenomenon before with the standard form $\int \frac{1}{x}\,dx = \log_e |x|$, where the logarithmic function was required for the integration of another purely algebraic function.

THE FUNCTIONS $y = \frac{1}{\sqrt{1-x^2}}$ AND $y = \frac{1}{1+x^2}$

The primitives of both these functions have now been obtained, and they should therefore be regarded as reasonably standard functions whose graphs should be known. The sketch of each function, and some important definite integrals associated with them, are developed in Questions 15 and 16 of the exercise.

TUTOR DEMO **Both forms of Box 4**

Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms given in Box 4.

YOUR TURN **Same method**

Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms given in Box 4.

WORKED EXAMPLE 6 **Keep for homework**

Evaluate $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$ using both standard forms given in Box 4.

SOLUTION

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx = \left[\sin^{-1} x\right]_0^{\frac{1}{2}}$$
$$= \sin^{-1}\tfrac{1}{2} - \sin^{-1} 0$$
$$= \tfrac{\pi}{6} - 0$$
$$= \tfrac{\pi}{6}$$

OR

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx = \left[-\cos^{-1} x\right]_0^{\frac{1}{2}}$$
$$= -\cos^{-1}\tfrac{1}{2} + \cos^{-1} 0$$
$$= -\tfrac{\pi}{3} + \tfrac{\pi}{2}$$
$$= \tfrac{\pi}{6}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 6

The exact values come from the special triangles: $\sin\frac{\pi}{6} = \frac{1}{2}$, $\sin 0 = 0$, $\cos\frac{\pi}{3} = \frac{1}{2}$ and $\cos\frac{\pi}{2} = 0$, and each angle lies in the range of its inverse function.

CONCEPT 2 · definite integrals with inverse tan

EXACT VALUES AND CALCULATOR VALUES

Enclose the primitive $\tan^{-1} x$ in square brackets, substitute the upper and lower limits, and subtract. When the limits give special angles, such as $\tan^{-1} 1 = \frac{\pi}{4}$ and $\tan^{-1} 0 = 0$, the answer is exact. Otherwise use the calculator, **in radians**, and round only at the end.

Because $1 + x^2$ is never zero, the function $\frac{1}{1+x^2}$ is continuous for all real x, so any limits may be used.

TUTOR DEMO **Exact or four significant figures**

Evaluate these definite integrals exactly or correct to four significant figures.

a $\int_0^{\sqrt{3}} \frac{1}{1+x^2}\,dx$

b $\int_0^{3} \frac{1}{1+x^2}\,dx$

YOUR TURN **Same method**

Evaluate these definite integrals exactly or correct to four significant figures.

a $\displaystyle\int_{-1}^{1} \frac{1}{1+x^2}\,dx$

b $\displaystyle\int_{0}^{2} \frac{1}{1+x^2}\,dx$

WORKED EXAMPLE 7 **Keep for homework**

Evaluate these definite integrals exactly or correct to four significant figures.

a $\displaystyle\int_{0}^{1} \frac{1}{1+x^2}\,dx$

b $\displaystyle\int_{0}^{4} \frac{1}{1+x^2}\,dx$

SOLUTION

a

$$\begin{aligned}\int_0^1 \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_0^1 \\ &= \tan^{-1} 1 - \tan^{-1} 0 \\ &= \tfrac{\pi}{4}\end{aligned}$$

b

$$\begin{aligned}\int_0^4 \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_0^4 \\ &= \tan^{-1} 4 - \tan^{-1} 0 \\ &\doteq 1.326\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 7

a $\tan^{-1} 1 - \tan^{-1} 0 = \frac{\pi}{4} - 0.$

b $\tan^{-1} 4 - \tan^{-1} 0 = 1.3258\ldots - 0.$

CONCEPT 3 · standard forms with one constant

MORE GENERAL STANDARD FORMS

When constants are involved, the calculation of the primitive becomes fiddly. These standard integrals are commonly used.

5 STANDARD FORMS WITH ONE CONSTANT

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a}+C \qquad \text{OR} \qquad \int \frac{1}{\sqrt{a^2-x^2}}\,dx = -\cos^{-1}\frac{x}{a}+C$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a}+C$$

PROOF

Let $u=\dfrac{x}{a}$. Then $\dfrac{du}{dx}=\dfrac{1}{a}$, and

$$\int \frac{1}{\sqrt{1-u^2}}\,\frac{du}{dx}\,dx = \sin^{-1}u \qquad \text{and} \qquad \int \frac{1}{1+u^2}\,\frac{du}{dx}\,dx = \tan^{-1}u.$$

A

$$\begin{aligned}\int \frac{1}{\sqrt{a^2-x^2}}\,dx &= \int \frac{1}{a\sqrt{1-\left(\frac{x}{a}\right)^2}}\,dx \\ &= \int \frac{1}{\sqrt{1-\left(\frac{x}{a}\right)^2}}\times\frac{1}{a}\,dx \\ &= \sin^{-1}\frac{x}{a}+C.\end{aligned}$$

B

$$\begin{aligned}\int \frac{1}{a^2+x^2}\,dx &= \int \frac{1}{a^2\left(1+\left(\frac{x}{a}\right)^2\right)}\,dx \\ &= \frac{1}{a}\int \frac{1}{1+\left(\frac{x}{a}\right)^2}\times\frac{1}{a}\,dx \\ &= \frac{1}{a}\tan^{-1}\frac{x}{a}+C.\end{aligned}$$

HOW THE PROOF WORKS

Here a is a positive constant. Both proofs take out the factor a^2, then use the reverse chain rule with $u = \dfrac{x}{a}$. By the chain rule, $\dfrac{d}{dx}\sin^{-1} u = \dfrac{1}{\sqrt{1-u^2}} \times \dfrac{du}{dx}$ and $\dfrac{d}{dx}\tan^{-1} u = \dfrac{1}{1+u^2} \times \dfrac{du}{dx}$. Reversing these gives the two integrals in the first line of the proof.

In **A**, $a^2 - x^2 = a^2\left(1 - \left(\frac{x}{a}\right)^2\right)$, and $\sqrt{a^2} = a$ because $a > 0$. In **B**, $a^2 + x^2 = a^2\left(1 + \left(\frac{x}{a}\right)^2\right)$.

In **B**, $\frac{1}{a^2}$ was written as $\frac{1}{a} \times \frac{1}{a}$: one $\frac{1}{a}$ is taken outside the integral, and the other is the $\frac{du}{dx}$ that the reverse chain rule needs. The $\cos^{-1}$ form follows in the same way from $\dfrac{d}{dx}\left(-\cos^{-1} u\right) = \dfrac{1}{\sqrt{1-u^2}} \times \dfrac{du}{dx}$.

READING OFF a

Match the integrand with the box: the number in front of x^2 must be 1. Then a^2 is the other number, and a is its positive square root. For example, $\sqrt{9 - x^2}$ has $a = 3$, and $8 + x^2$ has $a = \sqrt{8} = 2\sqrt{2}$. A constant factor in the numerator is taken outside the integral first.

TUTOR DEMO **Box 5 applied immediately**

Find these indefinite integrals.

a $\displaystyle\int \frac{3}{\sqrt{16 - x^2}}\,dx$

b $\displaystyle\int \frac{1}{12 + x^2}\,dx$

YOUR TURN **Same method**

Find these indefinite integrals.

a $\displaystyle\int \frac{5}{\sqrt{25-x^2}}\,dx$

b $\displaystyle\int \frac{1}{18+x^2}\,dx$

WORKED EXAMPLE 8A, B **Keep for homework**

Evaluate these indefinite integrals. In parts **a** and **b**, the formulae can be applied immediately.

a $\displaystyle\int \frac{2}{\sqrt{9-x^2}}\,dx$

b $\displaystyle\int \frac{1}{8+x^2}\,dx$

SOLUTION

a

$$\int \frac{2}{\sqrt{9-x^2}}\,dx = 2\sin^{-1}\frac{x}{3}+C$$

b

$$\int \frac{1}{8+x^2}\,dx = \frac{1}{2\sqrt{2}}\tan^{-1}\frac{x}{2\sqrt{2}}+C$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 8A, B

In **a**, $a=3$. In **b**, $a=\sqrt{8}=2\sqrt{2}$.

$$\int \frac{2}{\sqrt{9-x^2}}\,dx = 2\int \frac{1}{\sqrt{3^2-x^2}}\,dx, \qquad \int \frac{1}{8+x^2}\,dx = \int \frac{1}{(2\sqrt{2})^2+x^2}\,dx.$$

CONCEPT 4 · when x^2 has a coefficient

TAKE THE COEFFICIENT OF x^2 OUT FIRST

In integrals such as $\int \frac{6}{49+25x^2}\,dx$ and $\int \frac{1}{\sqrt{5-3x^2}}\,dx$, the coefficients of x^2 need to be taken out first, so that Box 5 can be used.

- For $a^2+b^2x^2$, take out b^2: $a^2+b^2x^2 = b^2\left(\frac{a^2}{b^2}+x^2\right)$.
- For $\sqrt{a^2-b^2x^2}$, take out $\sqrt{b^2}=b$:
$$\sqrt{a^2-b^2x^2} = \sqrt{b^2\left(\frac{a^2}{b^2}-x^2\right)} = b\sqrt{\frac{a^2}{b^2}-x^2}.$$

FULLER VERSIONS OF THE STANDARD FORMS

Because manipulating the constants is still difficult, some prefer to remember fuller versions of the standard forms. These forms can be proven in the same manner as the forms with a single constant, or they can be developed from those forms by taking out the coefficient of x^2.

DEVELOPING BOX 6 FROM BOX 5

Here a and b are positive constants. Take out the coefficient of x^2:

$$\begin{aligned}\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx &= \frac{1}{b}\int \frac{1}{\sqrt{\left(\frac{a}{b}\right)^2-x^2}}\,dx \\ &= \frac{1}{b}\sin^{-1}\left(x \div \frac{a}{b}\right)+C \\ &= \frac{1}{b}\sin^{-1}\frac{bx}{a}+C,\end{aligned}$$

$$\begin{aligned}\int \frac{1}{a^2+b^2x^2}\,dx &= \frac{1}{b^2}\int \frac{1}{\left(\frac{a}{b}\right)^2+x^2}\,dx \\ &= \frac{1}{b^2}\times\frac{b}{a}\tan^{-1}\left(x \div \frac{a}{b}\right)+C \\ &= \frac{1}{ab}\tan^{-1}\frac{bx}{a}+C.\end{aligned}$$

In the same way, the $\cos^{-1}$ form of Box 5 gives $-\frac{1}{b}\cos^{-1}\frac{bx}{a}+C$.

6 STANDARD FORMS WITH TWO CONSTANTS

$$\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx = \frac{1}{b}\sin^{-1}\frac{bx}{a}+C \qquad \text{OR} \qquad -\frac{1}{b}\cos^{-1}\frac{bx}{a}+C$$

$$\int \frac{1}{a^2+b^2x^2}\,dx = \frac{1}{ab}\tan^{-1}\frac{bx}{a}+C$$

With these more general forms, integrals like parts **c** and **d** of the worked example below can be written down without any intermediate working.

TUTOR DEMO **Coefficients of x^2**

Find these indefinite integrals by taking out the coefficient of x^2. Then check each answer with Box 6.

c $\displaystyle\int \frac{10}{9+16x^2}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{7-2x^2}}\,dx$

YOUR TURN **Same method**

Find these indefinite integrals by taking out the coefficient of x^2. Then check each answer with Box 6.

c $\displaystyle\int \frac{12}{25+4x^2}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{3-5x^2}}\,dx$

WORKED EXAMPLE 8C, D **Keep for homework**

Evaluate these indefinite integrals. In parts **c** and **d**, the coefficients of x^2 need to be taken out first.

c $\displaystyle\int \frac{6}{49+25x^2}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{5-3x^2}}\,dx$

SOLUTION

c

$$\begin{aligned}\int \frac{6}{49+25x^2}\,dx &= \frac{6}{25}\int \frac{1}{\frac{49}{25}+x^2}\,dx \\ &= \frac{6}{25}\times\frac{5}{7}\tan^{-1}\frac{x}{7/5}+C \\ &= \frac{6}{35}\tan^{-1}\frac{5x}{7}+C\end{aligned}$$

d

$$\begin{aligned}\int \frac{1}{\sqrt{5-3x^2}}\,dx &= \frac{1}{\sqrt{3}}\int \frac{1}{\sqrt{\frac{5}{3}-x^2}}\,dx \\ &= \tfrac{1}{3}\sqrt{3}\sin^{-1}x\sqrt{\tfrac{3}{5}}+C\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 8C, D

c Take out 25. Then $a^2=\frac{49}{25}$, so $a=\frac{7}{5}$. Here $\dfrac{1}{a}=\dfrac{5}{7}$, and $x\div\dfrac{7}{5}=\dfrac{5x}{7}$.

d Take out $\sqrt{3}$. Then $a^2=\frac{5}{3}$, so $a=\sqrt{\frac{5}{3}}$. Here $\sqrt{5-3x^2}=\sqrt{3}\sqrt{\frac{5}{3}-x^2}$, $\dfrac{1}{\sqrt{3}}=\dfrac{\sqrt{3}}{3}$, and $x\div\sqrt{\frac{5}{3}}=x\sqrt{\frac{3}{5}}$. The full step is

$$\frac{1}{\sqrt{3}}\int \frac{1}{\sqrt{\frac{5}{3}-x^2}}\,dx=\frac{1}{\sqrt{3}}\sin^{-1}\frac{x}{\sqrt{5/3}}+C.$$

Check with Box 6. In **c**, $a=7$ and $b=5$, so the answer is $6\times\dfrac{1}{35}\tan^{-1}\dfrac{5x}{7}+C$. In **d**, $a=\sqrt{5}$ and $b=\sqrt{3}$, so the answer is $\dfrac{1}{\sqrt{3}}\sin^{-1}\dfrac{\sqrt{3}\,x}{\sqrt{5}}+C$, which is the same.

CONCEPT 5 · the reverse chain rule

EXTENDING THE STANDARD FORMS

In the usual way, the standard forms can be extended to give forms appropriate for the reverse chain rule. These forms are the reversals of the chain-rule standard forms at the end of Lesson 12A.

BOX 3 FROM LESSON 12A

$$\frac{d}{dx}\sin^{-1}u=\frac{u'}{\sqrt{1-u^2}},\qquad \frac{d}{dx}\cos^{-1}u=\frac{-u'}{\sqrt{1-u^2}},\qquad \frac{d}{dx}\tan^{-1}u=\frac{u'}{1+u^2}.$$

7 THE REVERSE CHAIN RULE

$$\int\frac{u'}{\sqrt{1-u^2}}\,dx=\sin^{-1}u+C$$

$$\int\frac{u'}{\sqrt{1-u^2}}\,dx=-\cos^{-1}u+C$$

$$\int\frac{u'}{1+u^2}\,dx=\tan^{-1}u+C$$

OR

$$\int\frac{f'(x)}{\sqrt{1-\big(f(x)\big)^2}}\,dx=\sin^{-1}f(x)+C$$

$$\int\frac{f'(x)}{\sqrt{1-\big(f(x)\big)^2}}\,dx=-\cos^{-1}f(x)+C$$

$$\int\frac{f'(x)}{1+\big(f(x)\big)^2}\,dx=\tan^{-1}f(x)+C$$

FINDING u

Write the squared term as u^2, so that u is the quantity being squared, and find u'. The numerator must be u'. If it is only a constant multiple of u', adjust the constant outside the integral, as in Worked example 9b.

TUTOR DEMO **Find u and u'**

Find primitives of:

a $\dfrac{e^{2x}}{1+e^{4x}}$

b $\dfrac{x^2}{\sqrt{1-x^6}}$

YOUR TURN **Same method**

Find primitives of:

a $\dfrac{\cos x}{1+\sin^2 x}$

b $\dfrac{x}{1+x^4}$

WORKED EXAMPLE 9 **Keep for homework**

Find primitives of:

a $\dfrac{e^x}{1+e^{2x}}$ b $\dfrac{x}{\sqrt{1-x^4}}$

SOLUTION

a Let $u = e^x$, then $u' = e^x$. OR Let $f(x) = e^x$, then $f'(x) = e^x$.

With both notations,

$$\int \frac{e^x}{1+e^{2x}}\,dx = \int \frac{e^x}{1+\left(e^x\right)^2}\,dx$$
$$= \tan^{-1} e^x + C, \text{ for some constant } C.$$

b Let $u = x^2$, then $u' = 2x$. OR Let $f(x) = x^2$, then $f'(x) = 2x$.

With both notations,

$$\int \frac{x}{\sqrt{1-x^4}}\,dx = \tfrac{1}{2}\int \frac{2x}{\sqrt{1-\left(x^2\right)^2}}\,dx$$
$$= \tfrac{1}{2}\sin^{-1} x^2 + C, \text{ for some constant } C.$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 9

a Note that $e^{2x} = \left(e^x\right)^2$.

b Note that $x^4 = \left(x^2\right)^2$, and that $x = \frac{1}{2} \times 2x$.

CONCEPT 6 · given a derivative, find an integral

THE PRIMITIVES OF THE INVERSE TRIGONOMETRIC FUNCTIONS

As always, the result of a product-rule differentiation can be used to obtain an integral. In particular, this allows the primitives of the inverse trigonometric functions to be obtained.

THE METHOD

- Differentiate $x \times (\text{the inverse function})$ by the product rule. One term of the derivative is the inverse function itself.
- Integrate both sides of the derivative. The integral of a derivative is the function itself.
- Rearrange to make the integral of the inverse function the subject. The other integral left over is found by the reverse chain rule.

TUTOR DEMO **The primitive of $\cos^{-1} x$**

a Differentiate $x \cos^{-1} x$, and hence find a primitive of $\cos^{-1} x$.
b Find the area under the curve $y = \cos^{-1} x$ from $x = 0$ to $x = 1$.

YOUR TURN **Same method**

a Differentiate $x \sin^{-1} \dfrac{x}{2}$, and hence find a primitive of $\sin^{-1} \dfrac{x}{2}$.
b Find the area under the curve $y = \sin^{-1} \dfrac{x}{2}$ from $x = 0$ to $x = 2$.

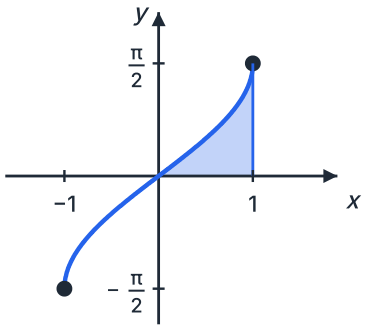
WORKED EXAMPLE 10 **Keep for homework**

a Differentiate $x\sin^{-1}x$, and hence find a primitive of $\sin^{-1}x$.
b Find the shaded area under the curve $y=\sin^{-1}x$ from $x=0$ to $x=1$.

SOLUTION

a Let $y=x\sin^{-1}x$.
Using the product rule with $u=x$ and $v=\sin^{-1}x$,

$$\frac{dy}{dx}=\sin^{-1}x+\frac{x}{\sqrt{1-x^2}}.$$



$$\text{Hence}\quad \int\sin^{-1}x\,dx+\int\frac{x}{\sqrt{1-x^2}}\,dx=x\sin^{-1}x$$

$$\int\sin^{-1}x\,dx=x\sin^{-1}x-\int\frac{x}{\sqrt{1-x^2}}\,dx.$$

Using the reverse chain rule,

$$-\int\frac{x}{\sqrt{1-x^2}}\,dx=\tfrac{1}{2}\int(1-x^2)^{-\frac{1}{2}}(-2x)\,dx$$
$$=\tfrac{1}{2}\times(1-x^2)^{\frac{1}{2}}\times\frac{2}{1}$$
$$=\sqrt{1-x^2},$$

Let $u=1-x^2$.
Then $\dfrac{du}{dx}=-2x$.
$\displaystyle\int u^{-\frac{1}{2}}\frac{du}{dx}\,dx=u^{\frac{1}{2}}\times 2.$

$$\text{so}\quad \int\sin^{-1}x\,dx=x\sin^{-1}x+\sqrt{1-x^2}+C.$$

b

$$\int_0^1\sin^{-1}x\,dx=\left[x\sin^{-1}x+\sqrt{1-x^2}\right]_0^1$$
$$=\left(1\times\tfrac{\pi}{2}+0\right)-(0+1)$$
$$=\tfrac{\pi}{2}-1\ \text{square units.}$$

Note: We have already established in Section 7D that the area under $y=\sin x$ from $x=0$ to $x=\frac{\pi}{2}$ is exactly 1 square unit. This means that the area between $y=\sin^{-1}x$ and the y-axis is 1, and subtracting this area from the rectangle of area $\frac{\pi}{2}$ in the diagram above gives the same value $\frac{\pi}{2}-1$ for the shaded area.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 10

a In the product rule, $u' = 1$ and $v' = \frac{1}{\sqrt{1-x^2}}$, so $\frac{dy}{dx} = 1 \times \sin^{-1} x + x \times \frac{1}{\sqrt{1-x^2}}$.

Integrating both sides: the integral of the derivative of $x \sin^{-1} x$ is $x \sin^{-1} x$. In the reverse chain rule, $-x = \frac{1}{2} \times (-2x)$, and $u^{-\frac{1}{2}}$ integrates to $u^{\frac{1}{2}} \div \frac{1}{2} = u^{\frac{1}{2}} \times 2$.

b The shaded area is $\int_0^1 \sin^{-1} x \, dx$.

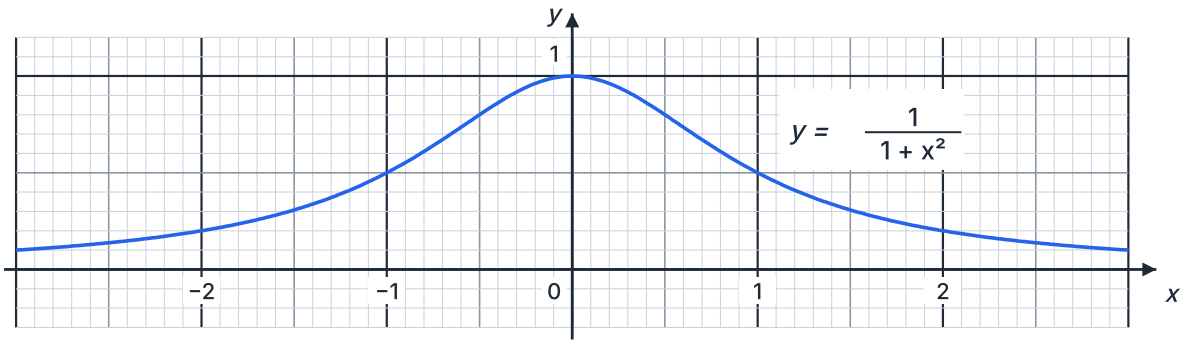
At $x = 1$: $\sin^{-1} 1 = \frac{\pi}{2}$ and $\sqrt{1-1} = 0$.

At $x = 0$: $0 \times \sin^{-1} 0 = 0$ and $\sqrt{1-0} = 1$.

TEXTBOOK HOMEWORK · Exercise 12B

FOUNDATION

1 a



Find each definite integral correct to two decimal places from the graph by counting the number of little squares in the region under the curve:

i $\int_0^1 \frac{1}{1+x^2} \, dx$

ii $\int_0^2 \frac{1}{1+x^2} \, dx$

iii $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{1+x^2} \, dx$

iv $\int_{-3}^{-1} \frac{1}{1+x^2} \, dx$

b Check your answers to **a** by using the fact that $\tan^{-1} x$ is a primitive of $\frac{1}{1+x^2}$.

2 Find:

a $\displaystyle\int \frac{-1}{\sqrt{1-x^2}}\,dx$ b $\displaystyle\int \frac{1}{\sqrt{4-x^2}}\,dx$ c $\displaystyle\int \frac{1}{9+x^2}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{\frac{4}{9}-x^2}}\,dx$ e $\displaystyle\int \frac{1}{2+x^2}\,dx$ f $\displaystyle\int \frac{-1}{\sqrt{5-x^2}}\,dx$

3 Find the exact value of:

a $\displaystyle\int_0^3 \frac{1}{\sqrt{9-x^2}}\,dx$ b $\displaystyle\int_0^2 \frac{1}{4+x^2}\,dx$ c $\displaystyle\int_0^1 \frac{1}{\sqrt{2-x^2}}\,dx$

d $\displaystyle\int_{\frac{1}{2}}^{\frac{1}{2}\sqrt{3}} \frac{\frac{1}{2}}{\frac{1}{4}+x^2}\,dx$ e $\displaystyle\int_{\frac{1}{6}\sqrt{3}}^{\frac{1}{6}} \frac{-1}{\sqrt{\frac{1}{9}-x^2}}\,dx$ f $\displaystyle\int_{-\frac{3}{4}\sqrt{2}}^{\frac{3}{4}} \frac{1}{\sqrt{\frac{9}{4}-x^2}}\,dx$

4 Find the equation of the curve, given that:

a $y' = \left(1-x^2\right)^{-\frac{1}{2}}$ and the curve passes through the point $(0, \pi)$.

b $y' = 4\left(16+x^2\right)^{-1}$ and the curve passes through the point $(-4, 0)$.

5 a If $y' = \dfrac{1}{\sqrt{36 - x^2}}$ and $y = \frac{\pi}{6}$ when $x = 3$, find the value of y when $x = 3\sqrt{3}$.

b Given that $y' = \dfrac{2}{4 + x^2}$ and that $y = \frac{\pi}{3}$ when $x = 2$, find y when $x = \frac{2}{\sqrt{3}}$.

DEVELOPMENT

6 Find:

a $\displaystyle\int \frac{1}{\sqrt{1 - 4x^2}}\,dx$

b $\displaystyle\int \frac{1}{1 + 16x^2}\,dx$

c $\displaystyle\int \frac{-1}{\sqrt{1 - 2x^2}}\,dx$

d $\displaystyle\int \frac{1}{\sqrt{4 - 9x^2}}\,dx$

e $\displaystyle\int \frac{1}{25 + 9x^2}\,dx$

f $\displaystyle\int \frac{-1}{\sqrt{3 - 4x^2}}\,dx$

7 Find the exact value of:

a $\displaystyle\int_0^{\frac{1}{6}} \frac{1}{\sqrt{1 - 9x^2}}\,dx$

b $\displaystyle\int_{\frac{1}{2}}^{\frac{1}{2}\sqrt{3}} \frac{2}{1 + 4x^2}\,dx$

c $\displaystyle\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1 - 3x^2}}\,dx$

d $\displaystyle\int_{-\frac{3}{4}}^{\frac{3}{2\sqrt{2}}} \frac{1}{\sqrt{9 - 4x^2}}\,dx$

8 a Shade the region bounded by $y = \sin^{-1} x$, the x-axis and the vertical line $x = \frac{1}{2}$.

b Show that $\frac{d}{dx}\left(x \sin^{-1} x + \sqrt{1 - x^2}\right) = \sin^{-1} x$.

c Hence find the exact area of the region.

9 a Shade the region bounded by the curve $y = \sin^{-1} x$, the y-axis and the line $y = \frac{\pi}{6}$.

b Find the exact area of this region.

c Hence use an alternative approach to confirm the area in the previous question.

10 a Show that $\frac{d}{dx}\left(\cos^{-1}(2 - x)\right) = \frac{1}{\sqrt{4x - x^2 - 3}}$.

b Hence find $\int_1^2 \frac{1}{\sqrt{4x - x^2 - 3}}\, dx$.

11 a Differentiate $\tan^{-1}\frac{1}{2}x^3$.

b Hence find $\displaystyle\int \frac{x^2}{4+x^6}\,dx$.

12 a Differentiate $x\tan^{-1}x$.

b Hence find $\displaystyle\int_0^1 \tan^{-1}x\,dx$.

13 Without finding any primitives, use arguments from symmetry or geometry to evaluate:

a $\displaystyle\int_{-\frac{1}{3}}^{\frac{1}{3}} \sin^{-1}x\,dx$

b $\displaystyle\int_{-5}^{5} \tan^{-1}x\,dx$

c $\displaystyle\int_{-\frac{3}{4}}^{\frac{3}{4}} \cos^{-1}x\,dx$

d $\displaystyle\int_{-\frac{2}{3}}^{\frac{2}{3}} \frac{x}{\sqrt{1-x^2}}\,dx$

e $\displaystyle\int_{-3}^{3} \frac{x}{1+x^2}\,dx$

f $\displaystyle\int_{-6}^{6} \sqrt{36-x^2}\,dx$

14 a Given that $f(x) = \dfrac{x}{1+x^2} - \tan^{-1} x$:

(i) find $f(0)$,

(ii) show that $f'(x) = \dfrac{-2x^2}{\left(1+x^2\right)^2}$.

b Hence:

(i) explain why $f(x) < 0$ for all $x > 0$,

(ii) find $\displaystyle\int_0^1 \frac{x^2}{\left(1+x^2\right)^2}\,dx$.

15 Consider the function $f(x) = \dfrac{1}{\sqrt{4 - x^2}}$.

a Sketch the graph of $y = \sqrt{4 - x^2}$.

b Hence sketch the graph of $y = f(x)$.

c Write down the domain and range of $f(x)$, and describe its symmetry.

d Find the area between the curve and the x-axis from $x = -1$ to $x = 1$.

e Find the total area between the curve and the x-axis. [**Note:** This is an example of an unbounded region having a finite area.]

16 Consider the function $f(x) = \dfrac{4}{x^2+4}$.

a What is the axis of symmetry of $y = f(x)$?

b What are the domain and range?

c Show that the graph of $f(x)$ has a maximum turning point at $(0, 1)$.

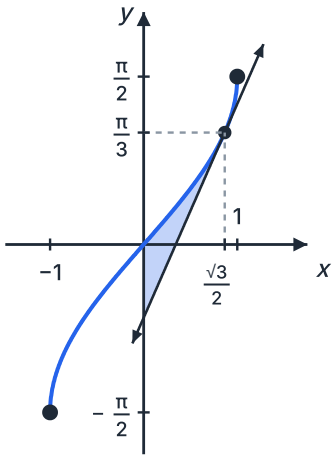
d Find $\lim\limits_{x \to \infty} f(x)$, and hence sketch $y = f(x)$. On the same axis, sketch $y = \frac{1}{4}\left(x^2+4\right)$.

e Calculate the area bounded by the curve and the x-axis from $x = -2\sqrt{3}$ to $x = \frac{2}{3}\sqrt{3}$.

f Find the exact area between the curve and the x-axis from $x = -a$ to $x = a$, where a is a positive constant.

g By letting a tend to infinity, find the total area between the curve and the x-axis. [**Note:** This is another example of an unbounded region having a finite area.]

17 The diagram to the right shows the region bounded by the curve $y = \sin^{-1} x$, the y-axis and the tangent to the curve at the point $\left(\frac{\sqrt{3}}{2}, \frac{\pi}{3}\right)$. Show that the region has area $\frac{1}{4}$ unit^2.



18

a Use the trapezoidal rule with five points to approximate $I = \int_0^1 \frac{1}{1+x^2}\,dx$, expressing your answer in simplest fraction form.

b Find the exact value of I, and hence show that $\pi \doteq \frac{5323}{1700}$.

19 Show that $\displaystyle\int_{-\frac{1}{4}}^{\frac{3}{5}} \frac{1}{1+x^2}\,dx = \frac{\pi}{4}$.

20 Find, using the reverse chain rule:

a $\displaystyle\int \frac{1}{\sqrt{x}\,(1+x)}\,dx$

b $\displaystyle\int_0^1 \frac{1}{e^{-x}+e^{x}}\,dx$

ENRICHMENT

21 a Show that $\dfrac{d}{dx}\left(\tan^{-1}\left(\frac{3}{2}\tan x\right)\right) = \dfrac{6}{5\sin^2 x + 4}$.

b Hence find the area under the curve $y = \dfrac{1}{5\sin^2 x + 4}$ from $x = 0$ to $x = 7$.

c Why is the calculation asked for in part **b** invalid and completely wrong?

22 [The power series for $\tan^{-1} x$]
Let x be a positive real number.

a Find the sum of the geometric series $1 - t^2 + t^4 - t^6 + \cdots + t^{4n}$, and hence show that for $0 < t < x$,

$$\frac{1}{1+t^2} < 1 - t^2 + t^4 - t^6 + \cdots + t^{4n}.$$

b Find $1 - t^2 + t^4 - t^6 + \cdots + t^{4n} - t^{4n+2}$, and hence show that for $0 < t < x$,

$$1 - t^2 + t^4 - t^6 + \cdots + t^{4n} < \frac{1}{1+t^2} + t^{4n+2}.$$

c By integrating the inequalities of parts **a** and **b** from $t = 0$ to $t = x$, show that

$$\tan^{-1} x < x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + \frac{x^{4n+1}}{4n+1} < \tan^{-1} x + \frac{x^{4n+3}}{4n+3}.$$

d By taking limits as $n \to \infty$, show that for $0 \leq x \leq 1$,

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots.$$

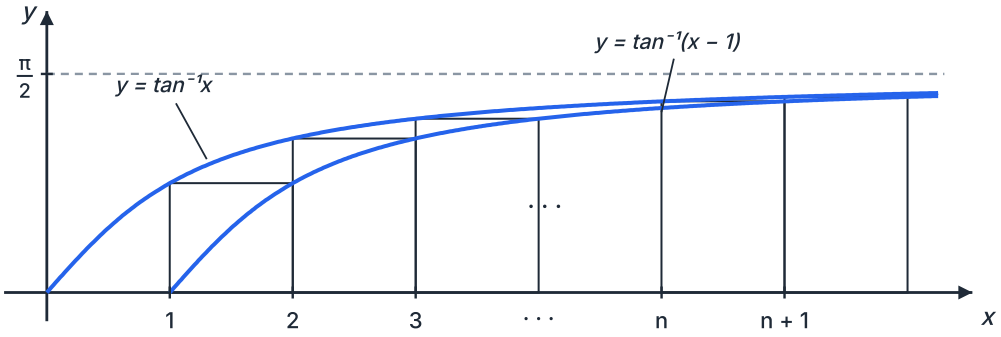
e Use the fact that $\tan^{-1} x$ is an odd function to prove this identity for $-1 \leq x < 0$.

f [Gregory's series] Use a suitable substitution to prove that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

g By combining the terms in pairs, show that $\frac{\pi}{8} = \frac{1}{1 \times 3} + \frac{1}{5 \times 7} + \frac{1}{9 \times 11} + \cdots$, and use the calculator to find how close an approximation to π can be obtained by taking 10 terms.

23 [A sandwiching argument]



In the diagram, n rectangles are constructed between the two curves $y = \tan^{-1} x$ and $y = \tan^{-1}(x-1)$ in the interval $1 \leq x \leq n+1$.

a Write down an expression for S_n, the sum of the areas of the n rectangles.

b Differentiate $x \tan^{-1} x$ and hence find a primitive of $\tan^{-1} x$.

c Show that for all $n \geq 1$,

$$n \tan^{-1} n - \tfrac{1}{2} \ln\left(n^2+1\right) < S_n$$
$$< (n+1)\tan^{-1}(n+1) - \tfrac{1}{2}\ln\left(\tfrac{n^2}{2} + n + 1\right) - \tfrac{\pi}{4}.$$

d Deduce that $1562 < \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 + \cdots + \tan^{-1} 1000 < 1565$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q5** **1 mark**

Which of the following is the correct expression for $\int \dfrac{dx}{\sqrt{16-x^2}}$?

A. $\sin^{-1} 4x + c$

B. $\cos^{-1} 4x + c$

C. $\sin^{-1} \dfrac{x}{4} + c$

D. $\cos^{-1} \dfrac{x}{4} + c$

Hint: how does the square root compare with $\sqrt{1-x^2}$?

PP 2 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q1** **1 mark**

Which is equal to $\int \dfrac{2}{2+x^2}\,dx$?

A. $\tan^{-1} \dfrac{x}{\sqrt{2}} + c$

B. $\dfrac{1}{\sqrt{2}} \tan^{-1} \dfrac{x}{\sqrt{2}} + c$

C. $\sqrt{2} \tan^{-1} \dfrac{x}{\sqrt{2}} + c$

D. $2 \tan^{-1} \dfrac{x}{\sqrt{2}} + c$

Hint: how does the denominator compare with $1 + x^2$?

PP 3 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q11(a)** **2 marks**

Find $\int \dfrac{5}{2+x^2}\,dx$.

Hint: which standard form has this shape?

PP 4 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9(b)** 2 marks

Evaluate

$$\int_0^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}}$$

Hint: how does the square root compare with $\sqrt{1-x^2}$?

PP 5 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q12(a)** 2 marks

Evaluate $\int_0^{\frac{1}{5}} \frac{dx}{1+25x^2}$

Hint: how does the denominator compare with $1+x^2$?

PP 6 **St George Girls High School, 2020, Trial HSC Examination, Mathematics Extension 1, Q15(a)** 2 marks

Find $\int_0^{0.125} \frac{2}{\sqrt{1-4x^2}}\,dx$. Write your answer correct to 3 significant figures.

Hint: how does the square root compare with $\sqrt{1-x^2}$?

PP 7 **Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(b)** 3 marks

Evaluate and find the exact value of $\int_{\sqrt{2}}^{\sqrt{6}} \frac{1}{2+x^2}\,dx$.

Hint: how does the denominator compare with $1+x^2$?

PP 8 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q7** 1 mark

What is the value of k such that $\int_0^k \frac{1}{\sqrt{4-x^2}}\,dx = \frac{\pi}{3}$?

A. 1

B. $\sqrt{3}$

C. 2

D. $2\sqrt{3}$

Hint: what is the integral in terms of k?

PP 9 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(a)** 2 marks

Find k such that $\int_0^k \frac{1}{\sqrt{9-x^2}}\,dx = \frac{\pi}{2}$

Hint: what is the primitive of the integrand?

PP 10 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9** 1 mark

What is the value of k such that $\int_0^k \frac{dx}{1+(x-1)^2} = \frac{\pi}{2}$

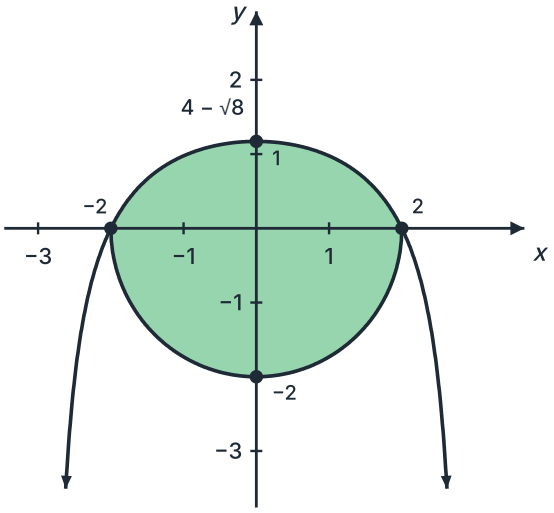
A. $2\sqrt{3}$ B. $\sqrt{3}$ C. 2 D. 1

Hint: how does the denominator compare with $1+x^2$?

PP 11 **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q14(b)** 3 marks

The region enclosed by $y = 4 - \frac{8}{\sqrt{8-x^2}}$ and the semi-circle $y = -\sqrt{4-x^2}$ is shaded in the diagram.

Find the exact value of the area of the shaded region.



Hint: which part of the region lies above the x-axis?

LESSON 12C

Further trigonometric integrals

The principal purpose of this section is the integration of $\sin^2 x$ and $\cos^2 x$. The keys are two forms of the $\cos 2\theta$ formulae: $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ and $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$. Trigonometric integrals in general are quickly reviewed, the squares of the other trigonometric functions are integrated using the Pythagorean identities, and the products-to-sums formulae turn products such as $\sin 3x \cos 2x$ into sums. Last, reverse-chain-rule integrations prepare for the next two sections on substitution.

CONCEPT 1 · six standard forms for integration

8 SIX STANDARD INTEGRALS

Reversing the derivatives of the six standard forms gives six standard integrals:

$$\int \cos x\,dx = \sin x + C \qquad \int \sin x\,dx = -\cos x + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad {}^*\!\int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

$${}^*\!\int \sec x \tan x\,dx = \sec x + C \qquad {}^*\!\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

The full list is there only for completeness, and so that the patterns can be seen. The three integrals marked * need not be memorised. They are the reversals of derivatives in Question 19 of Exercise 7B.

WHERE THE THREE STARRED INTEGRALS COME FROM

Write each reciprocal function in terms of $\sin x$ and $\cos x$, and differentiate with the chain rule or the quotient rule.

$$\begin{aligned}\frac{d}{dx}\sec x &= \frac{d}{dx}(\cos x)^{-1}\\ &= -(\cos x)^{-2} \times (-\sin x)\\ &= \frac{1}{\cos x} \times \frac{\sin x}{\cos x}\\ &= \sec x \tan x\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(-\operatorname{cosec} x) &= \frac{d}{dx}\big(-(\sin x)^{-1}\big)\\ &= (\sin x)^{-2} \times \cos x\\ &= \frac{1}{\sin x} \times \frac{\cos x}{\sin x}\\ &= \operatorname{cosec} x \cot x\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(-\cot x) = \frac{d}{dx}\left(-\frac{\cos x}{\sin x}\right) &= -\frac{(-\sin x)\sin x - \cos x \cos x}{\sin^2 x}\\ &= \frac{\sin^2 x + \cos^2 x}{\sin^2 x}\\ &= \frac{1}{\sin^2 x} = \operatorname{cosec}^2 x\end{aligned}$$

Reading each line backwards gives the three starred integrals.

A LINEAR FUNCTION OF x INSIDE (PART 2, LESSON 7D)

When the angle is $ax+b$, divide the standard primitive by a, the derivative of $ax+b$:

$$\int \cos(ax+b)\,dx = \tfrac{1}{a}\sin(ax+b)+C$$

$$\int \sin(ax+b)\,dx = -\tfrac{1}{a}\cos(ax+b)+C$$

$$\int \sec^2(ax+b)\,dx = \tfrac{1}{a}\tan(ax+b)+C$$

Check by differentiating: for example $\frac{d}{dx}\left(\frac{1}{a}\sin(ax+b)\right) = \frac{1}{a}\times a\cos(ax+b) = \cos(ax+b)$. Many integrals in this lesson use this rule.

TUTOR DEMO **Standard forms**

Find:

a $\displaystyle\int (2\cos x - \sin x)\,dx$

b $\displaystyle\int \sec^2 3x\,dx$

c $\displaystyle\int_0^{\frac{\pi}{3}} \sin\tfrac{1}{2}x\,dx$

YOUR TURN **Same skill**

Find:

a $\int (3\sin x + \cos x)\,dx$

b $\int \sec^2 \frac{1}{2}x\,dx$

c $\int_0^{\frac{\pi}{2}} \cos \frac{1}{3}x\,dx$

CONCEPT 2 · the primitives of $\sin^2 x$ and $\cos^2 x$

EXPRESSING $\sin^2\theta$ AND $\cos^2\theta$ IN TERMS OF $\cos 2\theta$ (SECTION 11A)

These two integrals are very important. The keys to finding them are two further forms of the $\cos 2\theta$ formulae.

WHERE THE TWO FORMS COME FROM

Start from $\cos 2\theta = \cos^2\theta - \sin^2\theta$. Using $\sin^2\theta = 1 - \cos^2\theta$, then $\cos^2\theta = 1 - \sin^2\theta$,

$$\cos 2\theta = 2\cos^2\theta - 1 \qquad \text{and} \qquad \cos 2\theta = 1 - 2\sin^2\theta.$$

From $\cos 2\theta = 2\cos^2\theta - 1$,

$2\cos^2\theta = 1 + \cos 2\theta$

$\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta.$

From $\cos 2\theta = 1 - 2\sin^2\theta$,

$2\sin^2\theta = 1 - \cos 2\theta$

$\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta.$

THE TWO IDENTITIES TO REMEMBER

$$\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta \qquad \cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$$

It is better to remember these identities rather than the actual integrals.

THE ANGLE DOUBLES

The square on the left becomes a cosine of the double angle on the right, and that can be integrated with the standard forms.

- Replacing θ by $3x$ gives $\sin^2 3x = \frac{1}{2} - \frac{1}{2}\cos 6x$, with $6x$, not $3x$. Replacing θ by $\frac{1}{2}x$ gives $\cos^2 \frac{1}{2}x = \frac{1}{2} + \frac{1}{2}\cos x$.
- Then divide by the coefficient of x in the new angle: $\int \cos 6x \, dx = \frac{1}{6}\sin 6x + C$.
- $\int \sin^2 x \, dx$ is not $\frac{1}{3}\sin^3 x + C$. Differentiating $\frac{1}{3}\sin^3 x$ gives $\sin^2 x \cos x$, not $\sin^2 x$.

HIGHER EVEN POWERS

For $\sin^4 x$ or $\cos^4 x$, write the power as a square of a square, for example $\cos^4 x = (\cos^2 x)^2$. Replace $\cos^2 x$ using the identity, expand, and use the identity again on the $\cos^2 2x$ term that appears (see Exercise 12C Question 10).

TUTOR DEMO **Squares of sine and cosine**

Evaluate:

a $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

b $\int_0^{\frac{2\pi}{3}} \cos^2 \frac{1}{2}x \, dx$

YOUR TURN **Same method**

Evaluate:

a $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$

b $\int_0^{\pi} \sin^2 \frac{1}{3}x \, dx$

WORKED EXAMPLE 11 **From the text: keep for homework**

Evaluate $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$.

SOLUTION

Using $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$,

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) dx \\ &= \left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} \\ &= \left(\frac{\pi}{4} - \frac{1}{4}\sin \pi\right) - \left(0 - \frac{1}{4}\sin 0\right) \\ &= \left(\frac{\pi}{4} - 0\right) - \left(0 - 0\right) \\ &= \frac{\pi}{4}\end{aligned}$$

CONCEPT 3 · the squares of all six trigonometric functions

THE SQUARES OF ALL SIX FUNCTIONS

Here are the primitives of the squares of all six trigonometric functions. Two are given as standard forms. The other four are given only in terms of the identities that are needed to obtain them.

THE IDENTITIES FOR $\tan^2 x$ AND $\cot^2 x$

For $\tan^2 x$ and $\cot^2 x$, divide $\sin^2 x + \cos^2 x = 1$ first by $\cos^2 x$ and then by $\sin^2 x$, as shown below.

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$
$$\tan^2 x + 1 = \sec^2 x$$
$$\tan^2 x = \sec^2 x - 1$$

$$\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} = \frac{1}{\sin^2 x}$$
$$1 + \cot^2 x = \operatorname{cosec}^2 x$$
$$\cot^2 x = \operatorname{cosec}^2 x - 1$$

9 INTEGRATING THE SQUARES OF THE TRIGONOMETRIC FUNCTIONS

$$\int \sin^2 x \, dx = \int \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 2x\right) dx \qquad \int \cos^2 x \, dx = \int \left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2x\right) dx$$

$$\int \sec^2 x \, dx = \tan x + C \qquad {}^*\int \operatorname{cosec}^2 x \, dx = -\cot x + C$$

$$\int \tan^2 x \, dx = \int \left(\sec^2 x - 1\right) dx \qquad {}^*\int \cot^2 x \, dx = \int \left(\operatorname{cosec}^2 x - 1\right) dx$$

The forms marked * are there for completeness.

TUTOR DEMO $\tan^2$ and $\cot^2$

a Find $\int \tan^2 3x \, dx$.

b Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^2 x \, dx$.

YOUR TURN Same method

a Find $\int \tan^2 \frac{1}{2}x \, dx$.

b Evaluate $\int_{0}^{\frac{\pi}{3}} \tan^2 x \, dx$.

CONCEPT 4 · products to sums (reviewed from Section 11A)

THE PRODUCTS-TO-SUMS FORMULAE

Exercise 12C Question 6 uses these formulae from Section 11A. They come from the compound-angle formulae

$$\sin(A+B) = \sin A\cos B + \cos A\sin B \quad \cos(A+B) = \cos A\cos B - \sin A\sin B$$
$$\sin(A-B) = \sin A\cos B - \cos A\sin B \quad \cos(A-B) = \cos A\cos B + \sin A\sin B$$

Adding and subtracting each pair, the other terms cancel:

$$\sin(A+B) + \sin(A-B) = 2\sin A\cos B$$
$$\sin(A+B) - \sin(A-B) = 2\cos A\sin B$$
$$\cos(A+B) + \cos(A-B) = 2\cos A\cos B$$
$$\cos(A+B) - \cos(A-B) = -2\sin A\sin B$$

PRODUCTS TO SUMS

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$
$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$
$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$
$$-2\sin A\sin B = \cos(A+B) - \cos(A-B)$$

A product of a sine and a cosine, two cosines, or two sines cannot be integrated directly. Rewritten as a sum, each term is a standard form with a linear angle.

TUTOR DEMO **Products to sums**

a Find $\int \sin 4x \cos x \, dx$.

b Evaluate $\int_0^{\frac{\pi}{6}} 2\sin 3x \sin x \, dx$.

YOUR TURN **Same method**

a Find $\int \cos 4x \sin 2x \, dx$.

b Evaluate $\int_0^{\frac{\pi}{4}} 2\cos 3x \cos x \, dx$.

CONCEPT 5 · trigonometric integrals using the reverse chain rule

TRIGONOMETRIC INTEGRALS USING THE REVERSE CHAIN RULE

It remains to demonstrate the use of the reverse chain rule in trigonometric integrals, in preparation for the next two sections on substitution.

THE REVERSE CHAIN RULE (PARTS 1 AND 2)

Let u be a function of x. The chain rule gives $\dfrac{d}{dx}\dfrac{u^{n+1}}{n+1} = u^n \dfrac{du}{dx}$ and $\dfrac{d}{dx}\log_e |u| = \dfrac{1}{u}\dfrac{du}{dx}$, so

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1) \qquad \text{and} \qquad \int \frac{u'}{u}\,dx = \log_e |u| + C.$$

Look for a function u whose derivative u' is also a factor of the integrand, apart from a constant. With trigonometric functions, $u = \sin x$ has $u' = \cos x$, and $u = \cos x$ has $u' = -\sin x$. Set out the working for u in a column on the right.

TUTOR DEMO **Reverse chain rule with $\log_e$**

Integrate $\tan 2x$ using the reverse chain rule.

YOUR TURN **Same method**

Integrate $\cot \frac{1}{2}x$ using the reverse chain rule.

WORKED EXAMPLE 12 **From the text: keep for homework**

Integrate $\cot x$ using the reverse chain rule.

SOLUTION

$$\begin{aligned}\int \cot x \, dx &= \int \frac{\cos x}{\sin x} \, dx \\ &= \log_e |\sin x| + C\end{aligned}$$

Let $u = \sin x$.

Then $u' = \cos x$,

and $\displaystyle\int \frac{u'}{u} \, dx = \log_e |u|$.

CONCEPT 6 · the reverse chain rule with powers of $\sin x$ and $\cos x$

POWERS OF $\sin x$ TIMES $\cos x$, AND POWERS OF $\cos x$ TIMES $\sin x$

- In $\cos x \sin^n x$, take $u = \sin x$. Then $\frac{du}{dx} = \cos x$ is already a factor, and the primitive is $\dfrac{u^{n+1}}{n+1}$.
- In $\sin x \cos^n x$, take $u = \cos x$. Then $\frac{du}{dx} = -\sin x$. Write $\sin x = -(-\sin x)$, so that the minus sign goes outside the integral.

TUTOR DEMO **Reverse chain rule with powers**

Use the reverse chain rule to find primitives of:

a $y = \sin x \cos^6 x$

b $y = \sec^2 x \tan^n x$

YOUR TURN **Same method**

Use the reverse chain rule to find primitives of:

a $y = \cos x \sin^5 x$

b $y = \sin x \cos^n x$

WORKED EXAMPLE 13 **From the text: keep for homework**

Use the reverse chain rule to find primitives of:
a $y = \sin x \cos^4 x$
b $y = \cos x \sin^n x$

SOLUTION

a

$$\begin{aligned}\int \sin x \cos^4 x \, dx &= -\int (-\sin x) \cos^4 x \, dx \\ &= -\tfrac{1}{5}\cos^5 x + C\end{aligned}$$

Let $u = \cos x$.

Then $\frac{du}{dx} = -\sin x$,

and $\int u^4 \frac{du}{dx} \, dx = \frac{1}{5}u^5$.

b

$$\int \cos x \sin^n x \, dx = \frac{\sin^{n+1} x}{n+1} + C$$

Let $u = \sin x$.

Then $\frac{du}{dx} = \cos x$,

and $\int u^n \frac{du}{dx} \, dx = \frac{u^{n+1}}{n+1}$.

TEXTBOOK HOMEWORK · Exercise 12C

FOUNDATION

1 Use double-angle results to show that:

a $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$

b $\cos^2 2x = \frac{1}{2} + \frac{1}{2}\cos 4x$

c $\sin 3x \cos 3x = \frac{1}{2}\sin 6x$

d $2\sin^2 \frac{x}{2} = 1 - \cos x$

2 Without a calculator, find the value of:

a $\cos^2 15°$

b $\sin^2 \frac{5\pi}{12}$

c $\sin 105° \cos 105°$

d $\sin^2 \frac{7\pi}{8}$

3 Express $\sin^2\theta$ in terms of $\cos 2\theta$, and hence find:

a $\int \sin^2 x \, dx$

b $\int \sin^2 2x \, dx$

c $\int \sin^2 \frac{1}{4}x \, dx$

d $\int \sin^2 3x \, dx$

4 Express $\cos^2\theta$ in terms of $\cos 2\theta$, and hence find:

a $\int \cos^2 x \, dx$

b $\int \cos^2 6x \, dx$

c $\int \cos^2 \frac{1}{2}x \, dx$

d $\int \cos^2 10x \, dx$

DEVELOPMENT

5 Use the results $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ and $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ to evaluate:

a $\int_0^{\pi} \sin^2 x \, dx$

b $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

c $\int_0^{\frac{\pi}{6}} \sin^2 \frac{1}{2}x \, dx$

d $\int_0^{\frac{\pi}{16}} \cos^2 2x \, dx$

e $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2\left(x + \frac{\pi}{12}\right) dx$

f $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2\left(x - \frac{\pi}{6}\right) dx$

6 Use the products-to-sums formulae reviewed in Section 11A to find:

a $\int \sin 3x \cos 2x \, dx$

b $\int \cos 3x \sin x \, dx$

c $\int_0^{\frac{\pi}{4}} 2 \cos 2x \cos x \, dx$

d $\int_0^{\frac{\pi}{3}} \sin 5x \sin 2x \, dx$

7 a Sketch the graph of $y = \cos 2x$, for $0 \leq x \leq 2\pi$.

b Hence sketch, on the same diagram, $y = \frac{1}{2}(1 + \cos 2x)$ and $y = \frac{1}{2}(1 - \cos 2x)$.

c Deduce graphically that $\cos^2 x + \sin^2 x = 1$.

8 Use the reverse chain rule to find:

a $\int \sin^3 x \cos x \, dx$ (Let $u = \sin x$.)

b $\int \sin^6 x \cos x \, dx$

c $\int \cos^5 x \sin x \, dx$ (Let $u = \cos x$.)

d $\int \cos^8 x \sin x \, dx$

e $\int e^x \sin e^x \, dx$ (Let $u = e^x$.)

f $\int e^x \cos 5e^x \, dx$

g $\int \tan x \, dx$ (Let $u = \cos x$.)

h $\int \cot 7x \, dx$

9 **a** Find the range of the function $y = \cos x \sin x$.

b Integrate $y = \cos x \sin x$ in three ways:
(i) using the $\sin 2x$ formula,
(ii) using the reverse chain rule in two different ways.

c Reconcile the three results.

10 a By writing $\sin^4 x$ as $(\sin^2 x)^2$, show that $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$.

b Find a similar result for $\cos^4 x$.

c Hence find:

(i) $\int_0^{\pi} \sin^4 x \, dx$

(ii) $\int_0^{\frac{\pi}{4}} \cos^4 x \, dx$

d Use the Pythagorean identity to find $\int_0^{\frac{\pi}{3}} \sin^3 x \, dx$.

11 Evaluate using the standard forms $\int \sec^2 x\,dx = \tan x$ and $\int \operatorname{cosec}^2 x\,dx = -\cot x$:

a $\int \tan^2 2x\,dx$

b $\int \cot^2 \frac{1}{2}x\,dx$

c $\int_{\frac{\pi}{12}}^{\frac{\pi}{9}} 3\tan^2 3x\,dx$

d $\int_{\frac{\pi}{24}}^{\frac{\pi}{8}} \cot^2 4x\,dx$

12 Integrate these functions, using the reverse chain rule or otherwise.

a $\tan x \sec^2 x$

b $\dfrac{\sin^2 x}{1+\cos x}$

c $\dfrac{1+\cos^3 x}{\cos^2 x}$

ENRICHMENT

13 Define $F(x) = \int_0^x \sin^2 t \, dt$, where $0 \leq x \leq 2\pi$.

a Show that $F(x) = \frac{1}{2}x - \frac{1}{4}\sin 2x$.

b Explain why $F'(x) = \sin^2 x$. Hence state the values of x in the given domain for which $F(x)$ is:

(i) stationary, (ii) increasing, (iii) decreasing.

c Explain why $F(x)$ never differs from $\frac{1}{2}x$ by more than $\frac{1}{4}$.

d Find any points of inflection of $F(x)$ in the given domain.

e Sketch, on the same diagram, the graphs of $y = F(x)$ and $y = F'(x)$ over the given domain, and observe how they are related.

f (i) For what value of k is $\int_0^k \sin^2 x \, dx = \frac{3\pi}{2}$?

(ii) For what values of k is $\int_0^k \sin^2 x \, dx = \frac{n\pi}{2}$, where n is an integer?

14 Find the value of $\lim_{R\to\infty}\left(\frac{1}{R}\int_0^R \sin^2 t\,dt\right)$, explaining your reasoning carefully.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q8** 1 mark

$\int \sin^2 3x\,dx$ is equal to which of the following?

(A) $\frac{x}{2}-\frac{\sin 6x}{3}+C$

(B) $\frac{x}{2}-\frac{\sin 6x}{6}+C$

(C) $\frac{x}{2}-\frac{\sin 6x}{9}+C$

(D) $\frac{x}{2}-\frac{\sin 6x}{12}+C$

Hint: which standard form is closest to this?

PP 2 **Normanhurst Boys High School, 2020, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q7** 1 mark

Which expression is equal to $\int \cos 3x \ \cos 2x \, dx$?

(A) $\frac{1}{10}\sin 5x + \frac{1}{2}\sin x + c$

(B) $\frac{1}{2}(\cos 5x + \cos x) + c$

(C) $\frac{1}{10}\cos 5x + \frac{1}{2}\cos x + c$

(D) $\frac{1}{2}(\sin 5x + \sin x) + c$

Hint: what could you simplify first?

PP 3 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q11(b)** 2 marks

Find $\int \sin^2 5x \, dx$.

Hint: what is the integrand in terms of the angle?

PP 4 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q11(a)** 3 marks

Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 2x \, dx$

Hint: what could you simplify before integrating?

PP 5 **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q11(c)** 3 marks

Find the area bounded by the $y = \cos^2(3x)$, the x-axis and the ordinates at $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$.

Hint: what does the region look like?

PP 6 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q12(a)** 3 marks

Find $\int (1 + \cos\theta)^2 \, d\theta$.

Hint: what could you simplify first?

PP 7 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q12(a)** 4 marks

(i) Show that $\sin(A+B)+\sin(A-B)=2\sin A\cos B$. 1

(ii) Hence, find $\int_0^{\frac{\pi}{2}} \sin 4x \cos 2x \, dx$ 3

Hint: what does each side of the identity in (i) contain?

PP 8 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q14(b)** 3 marks

Find $\int_0^{\frac{\pi}{8}} \cos^4 x \, dx$.

Hint: how could the integrand be written in a form you can integrate?

LESSON 12D

Integration by substitution

The reverse chain rule as we have been using it so far does not cover all the situations where the chain rule can be used in integration. This section and the next develop a more general method called *integration by substitution*. The first stage, covered in this lesson, begins by translating the reverse chain rule into a slightly more flexible notation. It involves substitutions of the form 'Let $u =$ some function of x.'

CONCEPT 1 · the reverse chain rule, an example

THE REVERSE CHAIN RULE: AN EXAMPLE

Here is an example of the reverse chain rule as we have been using it. The working is set out in full on the right.

TIP: THE REVERSE CHAIN RULE (PART 1, LESSON 5I)

Let u be a function of x, and let $F(u)$ be a primitive of $f(u)$, so that $F'(u) = f(u)$. The chain rule gives

$$\frac{d}{dx}F(u) = F'(u)\frac{du}{dx} = f(u)\frac{du}{dx}.$$

Reading this backwards gives the reverse chain rule. With $f(u) = u^n$, where $n \neq -1$, the primitive is $F(u) = \dfrac{u^{n+1}}{n+1}$:

$$\int f(u)\frac{du}{dx}\,dx = F(u) + C \qquad \text{and} \qquad \int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C.$$

Look for a function u whose derivative $\frac{du}{dx}$ is also a factor of the integrand, apart from a constant.

TUTOR DEMO **The reverse chain rule**

Find $\displaystyle\int x(3-x^2)^5\,dx$ using the reverse chain rule.

YOUR TURN **Same method**

Find $\int x(5-x^2)^3\,dx$ using the reverse chain rule.

WORKED EXAMPLE 14 **From the text: keep for homework**

Find $\int x(1-x^2)^4\,dx$.

SOLUTION

$$\begin{aligned}\int x(1-x^2)^4\,dx &= -\tfrac{1}{2}\int(-2x)(1-x^2)^4\,dx \\ &= -\tfrac{1}{2}\times\tfrac{1}{5}(1-x^2)^5+C \\ &= -\tfrac{1}{10}(1-x^2)^5+C\end{aligned}$$

Let $u=1-x^2$.

Then $\dfrac{du}{dx}=-2x$,

and $\displaystyle\int u^4\frac{du}{dx}\,dx=\tfrac{1}{5}u^5$.

CONCEPT 2 · rewriting the example as integration by substitution

REWRITING THIS EXAMPLE AS INTEGRATION BY SUBSTITUTION

We shall now rewrite this using a new notation. The key to this new notation is that the derivative $\dfrac{du}{dx}$ is treated as a fraction. The du and the dx are split apart, so that the statement

$$\frac{du}{dx}=-2x \qquad \textbf{is written instead as} \qquad du=-2x\,dx.$$

The new variable u no longer remains in the working column on the right, but is brought over into the main sequence of the solution on the left.

TIP: WHY THE NEW NOTATION IS VALID

Let $F(u)$ be a primitive of $f(u)$. In the variable u, $\int f(u)\,du = F(u) + C$. In the variable x, the reverse chain rule of Concept 1 gives $\int f(u)\,\dfrac{du}{dx}\,dx = F(u) + C$. Both integrals equal $F(u) + C$, so

$$\int f(u)\,\frac{du}{dx}\,dx = \int f(u)\,du.$$

So $\dfrac{du}{dx}\,dx$ may always be replaced by du, exactly as if $\dfrac{du}{dx}$ were a fraction and the two dx cancelled.

TIP: THE STEPS OF INTEGRATION BY SUBSTITUTION

1. Let $u =$ some function of x. Often it is the inside function, whose derivative is also a factor of the integrand.
2. Differentiate, and write the result with du and dx split apart, for example $du = -2x\,dx$.
3. Rearrange this so that it matches the rest of the integrand, for example $x\,dx = -\frac{1}{2}\,du$.
4. Rewrite the whole integral in terms of u and du. No x and no dx may remain.
5. Integrate with respect to u.
6. Replace u by its expression in x, and add the constant of integration.

TIP: COMMON ERRORS

- Keep the constant factor. From $du = -2x\,dx$, divide both sides by -2: $x\,dx = -\frac{1}{2}\,du$, not $-2\,du$.
- An integral such as $\int x\,u^4\,du$ is meaningless: the x and the dx must both be replaced before integrating.
- The answer to an indefinite integral in x must be given in terms of x again.

TUTOR DEMO **Substitution in an indefinite integral**

Find:

a $\int x(3-x^2)^5\,dx$, using the substitution $u = 3 - x^2$ (compare with the Concept 1 demo)

b $\int \sin x\sqrt{1+\cos x}\,dx$, using the substitution $u = 1 + \cos x$

YOUR TURN **Same method**

Find:

a $\int x(5-x^2)^3\,dx$, using the substitution $u = 5 - x^2$

b $\int \cos x\sqrt{3+\sin x}\,dx$, using the substitution $u = 3 + \sin x$

WORKED EXAMPLE 15 **From the text: keep for homework**

Find $\int x(1-x^2)^4\,dx$, using the substitution $u = 1 - x^2$.

SOLUTION

$$\begin{aligned}\int x(1-x^2)^4\,dx &= \int u^4\left(-\tfrac{1}{2}\right)du \\ &= -\tfrac{1}{2} \times \tfrac{1}{5}u^5 + C \\ &= -\tfrac{1}{10}(1-x^2)^5 + C\end{aligned}$$

Let $u = 1 - x^2$.

Then $du = -2x\,dx$,

and $x\,dx = -\tfrac{1}{2}\,du$.

WORKED EXAMPLE 16 **From the text: keep for homework**

Find $\int \sin x\sqrt{1-\cos x}\,dx$, using the substitution $u = 1 - \cos x$.

SOLUTION

$$\begin{aligned}\int \sin x\sqrt{1-\cos x}\,dx &= \int u^{\frac{1}{2}}\,du \\ &= \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{3}(1-\cos x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 1 - \cos x$.

Then $du = \sin x\,dx$.

TIP: THE PRIMITIVE IN EXAMPLE 16

The primitive of $u^{\frac{1}{2}}$ is $\dfrac{u^{\frac{3}{2}}}{\frac{3}{2}} = \tfrac{2}{3}u^{\frac{3}{2}}$.

CONCEPT 3 · an advance on the reverse chain rule

AN ADVANCE ON THE REVERSE CHAIN RULE

Some integrals that can be done in this way could only be done by the reverse chain rule in a rather clumsy manner.

TIP: WRITING THE LEFTOVER x IN TERMS OF u

In $\int x\sqrt{1-x}\,dx$, the derivative of the inside function $1-x$ is -1, so the factor x is not its derivative.

The substitution $u = 1 - x$ deals with this. It gives $du = -dx$, and the leftover x is written in terms of u by rearranging $u = 1 - x$:

$x = 1 - u.$

The integrand then becomes $(1-u)\sqrt{u}$, which expands into a sum of powers of u.

TIP: EXPANDING WITH THE INDEX LAWS

Write $\sqrt{u} = u^{\frac{1}{2}}$. Then $u\sqrt{u} = u^1 \times u^{\frac{1}{2}} = u^{\frac{3}{2}}$, and

$(1-u)\sqrt{u} = u^{\frac{1}{2}} - u^{\frac{3}{2}} \qquad \textbf{and} \qquad (u-1)\sqrt{u} = u^{\frac{3}{2}} - u^{\frac{1}{2}}.$

Each power is then integrated with $\int u^n\,du = \dfrac{u^{n+1}}{n+1} + C$, for example

$$\int u^{\frac{3}{2}}\,du = \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + C = \tfrac{2}{5}u^{\frac{5}{2}} + C.$$

TUTOR DEMO **Writing x in terms of u**

Find $\int x\sqrt{2-x}\,dx$, using the substitution $u = 2 - x$.

YOUR TURN **Same method**

Find $\int x\sqrt{x+3}\,dx$, using the substitution $u = x + 3$.

WORKED EXAMPLE 17 **From the text: keep for homework**

Find $\int x\sqrt{1-x}\,dx$, using the substitution $u = 1 - x$.

SOLUTION

$$\begin{aligned}\int x\sqrt{1-x}\,dx &= \int (1-u)\sqrt{u}\,(-du)\\ &= \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) du\\ &= \tfrac{2}{5}u^{\frac{5}{2}} - \tfrac{2}{3}u^{\frac{3}{2}} + C\\ &= \tfrac{2}{5}(1-x)^{\frac{5}{2}} - \tfrac{2}{3}(1-x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 1 - x$.

Then $du = -dx$,

and $x = 1 - u$.

TIP: THE SECOND LINE OF EXAMPLE 17

The second line takes the minus sign inside: $-(1-u)\sqrt{u} = (u-1)\sqrt{u} = u^{\frac{3}{2}} - u^{\frac{1}{2}}$.

CONCEPT 4 · substituting the limits of integration in a definite integral

SUBSTITUTING THE LIMITS OF INTEGRATION IN A DEFINITE INTEGRAL

A great advantage of this new method is that the limits of integration can be changed from values of x to values of u. There is then no need ever to go back to x. The first worked example below repeats the previous integrand, but this time within a definite integral.

TIP: FINDING THE NEW LIMITS

- Put each x-limit into the substitution to find the matching u-limit, and record this in the working column: 'When $x = a$, $u = \ldots$'.
- The u-value that comes from the lower x-limit is the lower u-limit, even when it is the larger number. Example 18 below gives $\int_1^0$.
- Evaluate the u-integral between the u-limits.

TIP: REVERSING THE LIMITS (PART 1, LESSON 5C)

Reversing the limits of an integral changes its sign:

$$\int_b^a f(u)\,du = -\int_a^b f(u)\,du.$$

So a minus sign in front of an integral can be removed by reversing its limits, for example $-\int_1^0 u^5\,du = \int_0^1 u^5\,du$. Either form may be used, and both give the same value.

TIP: NEVER MIX THE VARIABLES

- The limits of an integral with du must be values of u. Keeping the x-limits on a u-integral is wrong.
- Do not substitute back to x and then use the u-limits.

TUTOR DEMO **Definite integrals by substitution**

Find:

a $\int_0^2 x\sqrt{2-x}\,dx$, using the substitution $u = 2 - x$

b $\int_0^{\pi} \sin x \cos^4 x\,dx$, using the substitution $u = \cos x$

YOUR TURN **Same method**

Find:

a $\int_{-3}^1 x\sqrt{x+3}\,dx$, using the substitution $u = x + 3$

b $\int_0^{\frac{\pi}{2}} \sin x \cos^5 x\,dx$, using the substitution $u = \cos x$

WORKED EXAMPLE 18 **From the text: keep for homework**

Find $\int_0^1 x\sqrt{1-x}\,dx$, using the substitution $u = 1 - x$.

SOLUTION

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= -\int_1^0 (1-u)\sqrt{u}\,du \\ &= -\int_1^0 \left(u^{\frac{1}{2}} - u^{\frac{3}{2}}\right) du \\ &= -\left[\tfrac{2}{3}u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_1^0 \\ &= -0 + \left(\tfrac{2}{3} - \tfrac{2}{5}\right) \\ &= \tfrac{4}{15}\end{aligned}$$

Let $u = 1 - x$.

Then $du = -dx$,

and $x = 1 - u$.

When $x = 0$, $u = 1$,

when $x = 1$, $u = 0$.

TIP: THE ARITHMETIC IN EXAMPLE 18

The bracket is the value at the upper limit minus the value at the lower limit:

$$-\left[\tfrac{2}{3}u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_1^0 = -\left((0-0) - \left(\tfrac{2}{3} - \tfrac{2}{5}\right)\right) = -0 + \left(\tfrac{2}{3} - \tfrac{2}{5}\right)$$

At $u = 1$, the primitive is $\frac{2}{3} \times 1^{\frac{3}{2}} - \frac{2}{5} \times 1^{\frac{5}{2}} = \frac{2}{3} - \frac{2}{5}$, and $\frac{2}{3} - \frac{2}{5} = \frac{10}{15} - \frac{6}{15} = \frac{4}{15}$.

WORKED EXAMPLE 19 **From the text: keep for homework**

Find $\int_0^\pi \sin x \cos^6 x\,dx$, using the substitution $u = \cos x$.

SOLUTION

$$\begin{aligned}\int_0^\pi \sin x \cos^6 x\,dx &= -\int_1^{-1} u^6\,du \\ &= -\tfrac{1}{7}\left[u^7\right]_1^{-1} \\ &= -\tfrac{1}{7} \times (-1) + \tfrac{1}{7} \times 1 \\ &= \tfrac{2}{7}\end{aligned}$$

Let $u = \cos x$.

Then $du = -\sin x\,dx$.

When $x = 0$, $u = 1$,

when $x = \pi$, $u = -1$.

TIP: THE BRACKET IN EXAMPLE 19

The bracket is the value at the upper limit minus the value at the lower limit:

$$-\tfrac{1}{7}\left[u^7\right]_1^{-1} = -\tfrac{1}{7}\left((-1)^7 - 1^7\right) = -\tfrac{1}{7} \times (-1) + \tfrac{1}{7} \times 1$$

TEXTBOOK HOMEWORK · Exercise 12D

FOUNDATION

1 Consider the integral $\int 2x(1+x^2)^3\,dx$, and the substitution $u = 1 + x^2$.

a Show that $du = 2x\,dx$.

b Show that the integral can be written as $\int u^3\,du$.

c Hence find the primitive of $2x(1+x^2)^3$.

d Check your answer by differentiating it.

2 Repeat the previous question for each indefinite integral and substitution.

a $\int 2(2x+3)^3\,dx$ (Let $u = 2x+3$.)

b $\int 3x^2(1+x^3)^4\,dx$ (Let $u = 1+x^3$.)

c $\int \dfrac{2x}{(1+x^2)^2}\,dx$ (Let $u = 1+x^2$.)

d $\int \dfrac{3}{\sqrt{3x-5}}\,dx$ (Let $u = 3x-5$.)

e $\int \sin^3 x \cos x\,dx$ (Let $u = \sin x$.)

f $\int \dfrac{4x^3}{1+x^4}\,dx$ (Let $u = 1+x^4$.)

3 Consider the integral $\int \dfrac{x}{\sqrt{1-x^2}}\,dx$, and the substitution $u = 1 - x^2$.

a Show that $x\,dx = -\frac{1}{2}\,du$.

b Show that the integral can be written as $-\frac{1}{2}\int u^{-\frac{1}{2}}\,du$.

c Hence find the primitive of $\dfrac{x}{\sqrt{1-x^2}}$.

4 Repeat the previous question for each indefinite integral and substitution.

a $\int x^3(x^4+1)^5\,dx$ (Let $u = x^4+1$.)

b $\int x^2\sqrt{x^3-1}\,dx$ (Let $u = x^3-1$.)

c $\int x^2 e^{x^3}\,dx$ (Let $u = x^3$.)

d $\int \dfrac{1}{\sqrt{x}(1+\sqrt{x})^3}\,dx$ (Let $u = 1+\sqrt{x}$.)

e $\int \tan^2 2x \sec^2 2x\,dx$ (Let $u = \tan 2x$.)

f $\int \dfrac{e^{\frac{1}{x}}}{x^2}\,dx$ (Let $u = \dfrac{1}{x}$.)

5 Find the exact value of each definite integral, using the given substitution.

a $\displaystyle\int_0^1 x^2(2+x^3)^3\,dx$ (Let $u = 2 + x^3$.)

b $\displaystyle\int_0^1 \frac{2x^3}{\sqrt{1+x^4}}\,dx$ (Let $u = 1 + x^4$.)

c $\displaystyle\int_0^{\frac{\pi}{2}} \cos^2 x \sin x\,dx$ (Let $u = \cos x$.)

d $\displaystyle\int_{\frac{\sqrt{3}}{2}}^1 x\sqrt{1-x^2}\,dx$ (Let $u = 1 - x^2$.)

e $\displaystyle\int_1^{e^2} \frac{\ln x}{x}\,dx$ (Let $u = \ln x$.)

f $\displaystyle\int_0^4 \frac{e^{\sqrt{x}}}{4\sqrt{x}}\,dx$ (Let $u = \sqrt{x}$.)

g $\displaystyle\int_0^{\frac{\pi}{4}} \sin^4 2x \cos 2x\,dx$ (Let $u = \sin 2x$.)

h $\displaystyle\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}}\,dx$ (Let $u = \sin^{-1} x$.)

i $\displaystyle\int_0^2 \frac{x+1}{\sqrt[3]{x^2+2x}}\,dx$ (Let $u = x^2 + 2x$.)

j $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan x}\,dx$ (Let $u = \tan x$.)

DEVELOPMENT

6 Use the substitution $u = x^3$ to find the exact area bounded by the curve $y = \dfrac{x^2}{1+x^6}$, the x-axis and the line $x = 1$.

7 Evaluate each definite integral, using the substitution $u = \sin x$.

a $\displaystyle\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\,dx$

b $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin^2 x}\,dx$

c $\displaystyle\int_0^{\frac{\pi}{2}} \cos^3 x\,dx$

d $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin^4 x}\,dx$

8 Find each indefinite integral, using the given substitution.

a $\displaystyle\int \frac{e^{2x}}{\sqrt{1+e^{2x}}}\,dx$ (Let $u = e^{2x}$.)

b $\displaystyle\int \frac{1}{x \ln x}\,dx$ (Let $u = \ln x$.)

c $\displaystyle\int \frac{\tan x}{\ln \cos x}\,dx$ (Let $u = \ln \cos x$.)

d $\displaystyle\int \tan^3 x \sec^4 x\,dx$ (Let $u = \tan x$.)

9

a A curve has gradient function $\dfrac{e^{2x}}{1+e^{4x}}$ and passes through the point $\left(0, \frac{\pi}{8}\right)$. Use the substitution $u = e^{2x}$ to find its equation.

b If $y'' = \dfrac{x}{(4-x^2)^{\frac{3}{2}}}$, and when $x = 0$, $y' = 1$ and $y = \frac{1}{2}$, use the substitution $u = 4 - x^2$ to find y' and then find y as a function of x.

10 a Show that $\dfrac{d}{dx}(\sec x) = \sec x \tan x$.

b Use the substitution $u = \sec x$, and the standard form $\displaystyle\int a^x \, dx = \frac{a^x}{\ln a}$, to find:

(i) $\displaystyle\int_0^{\frac{\pi}{3}} 2^{\sec x} \sec x \tan x \, dx$

(ii) $\displaystyle\int_0^{\frac{\pi}{4}} \sec^5 x \tan x \, dx$

11 Evaluate each integral, using the given substitution.

a $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{1 + \sin^2 x} \, dx$ (Let $u = \sin^2 x$.)

b $\displaystyle\int_1^e \frac{\ln x + 1}{(x \ln x + 1)^2} \, dx$ (Let $u = x \ln x$.)

12 Use the substitution $u = \sqrt{x-1}$ to find $\displaystyle\int \frac{1}{2x\sqrt{x-1}}\,dx$.

13 a Use the substitution $u = \sqrt{x}$ to find $\displaystyle\int \frac{1}{\sqrt{x(1-x)}}\,dx$.

b Evaluate the integral in part **a** again, using the substitution $u = x - \frac{1}{2}$.

c Hence show that $\sin^{-1}(2x-1) = 2\sin^{-1}\sqrt{x} - \frac{\pi}{2}$, for $0 < x < 1$.

ENRICHMENT

14 Use the substitution $u = x - \dfrac{1}{x}$ to show that $\displaystyle\int_1^{\frac{1}{2}(\sqrt{6}+\sqrt{2})} \frac{1+x^2}{1+x^4}\,dx = \frac{\pi}{4\sqrt{2}}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q8** 1 mark

By letting $u = \tan x$, $\displaystyle\int_0^{\frac{\pi}{3}} \tan^2 x \sec^2 x\,dx$ can be expressed as

(A) $\displaystyle\int_0^{\frac{\pi}{3}} u^2\,du$

(B) $\displaystyle\int_0^{\sqrt{3}} u^2\,du$

(C) $\displaystyle\int_0^{\sqrt{3}} u\,du$

(D) $\displaystyle\int_0^{\sqrt{3}} (u^4+u^2)\,du$

Hint: what does each part of the integral look like?

PP 2 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10(a)** 2 marks

Use the substitution $u = x^2 + 1$, or otherwise, to find

$$\int x\sqrt{x^2+1}\,dx.$$

Hint: how are the parts of the integrand related?

PP 3 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q11(d)** 2 marks

Use the substitution $u = 2x - 1$ to find $\int 4x\sqrt{2x-1}\,dx$.

Hint: what does the integral look like?

PP 4 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)** 3 marks

Use the substitution $u = \ln 3x$, to find $\int \dfrac{dx}{x(\ln 3x)^2}$

Hint: how is the integrand related to u?

PP 5 **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q11(b)** 3 marks

Use the substitution $u = 6 - x$ to find the exact value of $\int_1^6 x\sqrt{6-x}\,dx$

Hint: what does each part of the integral look like?

PP 6 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q12(a)** 3 marks

Find $\int \dfrac{\cos^2(\ln x)}{x}\,dx$ using the substitution $u = \ln x$.

Hint: what does the integrand contain?

PP 7 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d)** 3 marks

Use the substitution $u = \tan x - x$ to find $\int \tan^2 x \sqrt{\tan x - x}\, dx$.

Hint: how are the parts of the integrand related to u?

PP 8 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q12(b)** 3 marks

Use the substitution $u = 2e^x$ to show that

$$\int_{\ln\left(\frac{1}{2}\right)}^{\ln\left(\frac{\sqrt{3}}{2}\right)} \frac{e^x}{1+4e^{2x}}\, dx = \frac{\pi}{24}$$

Hint: how is each part of the integrand related to u?

LESSON 12E

Further integration by substitution

The second stage of integration by substitution reverses the procedure of Lesson 12D and replaces x by a function of u. The substitutions are therefore of the form 'Let $x =$ some function of u.' A substitution such as $x = 1 - u^2$ removes a square root, and a trigonometric substitution such as $x = 6\sin u$ turns $\sqrt{36 - x^2}$ into $6\cos u$. In a definite integral the limits are changed as well, so there is no need to return to x.

CONCEPT 1 · substituting x by a function of u

SUBSTITUTING x BY A FUNCTION OF u

The second stage of integration by substitution reverses the procedure of Lesson 12D and replaces x by a function of u. The substitutions are therefore of the form 'Let $x =$ some function of u.'

As a first example, here is a quite different substitution which solves the integral given in a worked example of the last section.

THE METHOD, STEP BY STEP

The earlier example is Lesson 12D Example 18, $\displaystyle\int_0^1 x\sqrt{1-x}\,dx$, where the substitution was $u = 1 - x$.

Write the substitution as $x = g(u)$. Then:

- Differentiate: $\dfrac{dx}{du} = g'(u)$, so $dx = g'(u)\,du$. The dx in the integral must be replaced by $g'(u)\,du$.
- Replace every x in the integrand by $g(u)$. The function g is chosen so that the awkward part of the integrand, usually a square root, becomes a simple expression in u.
- In a definite integral, change the limits from values of x to values of u. Then there is no need to return to x.
- In an indefinite integral, the answer must be written in terms of x at the end. Use the substitution backwards, for example $u = \sqrt{1-x}$ when $x = 1 - u^2$.

THE SUBSTITUTION $x = 1 - u^2$ IN DETAIL

Let $x = 1 - u^2$. Then

$$\frac{dx}{du} = -2u$$
$$dx = -2u\,du$$

$$\begin{aligned} 1 - x &= 1 - (1 - u^2) \\ &= u^2 \\ \sqrt{1 - x} &= u \end{aligned}$$

The positive square root $u \geq 0$ is taken, so $u = \sqrt{1 - x}$. For the limits: when $x = 0$, $1 - u^2 = 0$, so $u^2 = 1$ and $u = 1$; when $x = 1$, $1 - u^2 = 1$, so $u^2 = 0$ and $u = 0$.

CHANGE EVERYTHING

- Never just write du in place of dx. Here $dx = -2u\,du$, and the factor $-2u$ stays in the integral.
- The new limits may come out in the reverse order (here from $u = 1$ to $u = 0$). Keep them in that order, and take the minus sign outside if it helps.

TUTOR DEMO **Let x be a function of u**

a Find $\int x\sqrt{x - 2}\,dx$, using the substitution $x = u^2 + 2$.

b Find $\int_{-3}^{1} x\sqrt{1 - x}\,dx$, using the substitution $x = 1 - u^2$.

YOUR TURN **Same method**

a Find $\int x\sqrt{x+3}\,dx$, using the substitution $x = u^2 - 3$.

b Find $\int_{-3}^{0} x\sqrt{1-x}\,dx$, using the substitution $x = 1 - u^2$.

WORKED EXAMPLE 20 **From the text: keep for homework**

Find $\int_0^1 x\sqrt{1-x}\,dx$, using the substitution $x = 1 - u^2$.

SOLUTION

$$\begin{aligned}\int_0^1 x\sqrt{1-x}\,dx &= \int_1^0 (1-u^2)\,u\,(-2u)\,du \\ &= -2\int_1^0 (u^2 - u^4)\,du \\ &= -2\left[\tfrac{1}{3}u^3 - \tfrac{1}{5}u^5\right]_1^0 \\ &= -2\left(0 - \left(\tfrac{1}{3} - \tfrac{1}{5}\right)\right) \\ &= 2\left(\tfrac{1}{3} - \tfrac{1}{5}\right) \\ &= 2 \times \tfrac{2}{15} \\ &= \tfrac{4}{15}\end{aligned}$$

Let $x = 1 - u^2$.

Then $dx = -2u\,du$,

and $\sqrt{1-x} = u$.

When $x = 0$, $u = 1$,

when $x = 1$, $u = 0$.

This question is a good example of how an integral may be evaluated in contrasting ways.

READING WORKED EXAMPLE 20

- In the second line, $(1-u^2)\times u\times(-2u) = -2(u^2-u^4)$.
- At the upper limit $u=0$ the bracket is 0, and at the lower limit $u=1$ it is $\frac{1}{3}-\frac{1}{5}$.
- Lesson 12D found the same value, $\frac{4}{15}$, with the substitution $u=1-x$.

CONCEPT 2 · a trigonometric substitution, and the area of a segment

A TRIGONOMETRIC SUBSTITUTION

The next integral uses a trigonometric substitution, but can also be done using areas of segments.

WHY $x=a\sin u$ REMOVES $\sqrt{a^2-x^2}$

When the integrand contains $\sqrt{a^2-x^2}$, where $a>0$, let $x=a\sin u$. Then $dx=a\cos u\,du$, and

$$\begin{aligned} a^2-x^2 &= a^2-a^2\sin^2 u \\ &= a^2(1-\sin^2 u) \\ &= a^2\cos^2 u, \end{aligned}$$

so $\sqrt{a^2-x^2}=a\cos u$, because $\cos u\geq 0$ when $-\frac{\pi}{2}\leq u\leq\frac{\pi}{2}$. The integral usually becomes a multiple of $\cos^2 u$, which is integrated with $\cos^2 u=\frac{1}{2}+\frac{1}{2}\cos 2u$ (Lesson 12C).

THE AREA OF A SEGMENT (YEAR 11)

A chord of a circle of radius r that subtends an angle θ (in radians) at the centre cuts off a segment. The segment is the sector minus the isosceles triangle formed by the two radii and the chord:

$$\text{segment}=\tfrac{1}{2}r^2\theta-\tfrac{1}{2}r^2\sin\theta=\tfrac{1}{2}r^2(\theta-\sin\theta).$$

The sector has area $\frac{1}{2}r^2\theta$. The triangle has two sides r with the angle θ between them, so its area is $\frac{1}{2}r\times r\times\sin\theta$.

TUTOR DEMO **Two ways**

Find $\int_{2\sqrt{2}}^{4} \sqrt{16-x^2}\,dx$:

a using the substitution $x = 4\sin u$,

b using the formula for the area of a segment.

YOUR TURN **Same methods**

Find $\int_{0}^{3} \sqrt{36-x^2}\,dx$:

a using the substitution $x = 6\sin u$,

b using the areas of a triangle and a sector.

WORKED EXAMPLE 21 **From the text: keep for homework**

Find $\int_{3\sqrt{2}}^{6} \sqrt{36-x^2}\,dx$:

a using the substitution $x = 6\sin u$,

b using the formula for the area of a segment.

SOLUTION

a

$$\begin{aligned}
\int_{3\sqrt{2}}^{6} \sqrt{36-x^2}\,dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 6\cos u \times 6\cos u\,du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 36\cos^2 u\,du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 36\left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2u\right)du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (18+18\cos 2u)\,du \\
&= \left[18u+9\sin 2u\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
&= \left(9\pi+9\sin\pi\right)-\left(\tfrac{9\pi}{2}+9\sin\tfrac{\pi}{2}\right) \\
&= (9\pi+0)-\left(\tfrac{9\pi}{2}+9\right) \\
&= \tfrac{9}{2}(\pi-2)
\end{aligned}$$

Let $x = 6\sin u$.

Then $dx = 6\cos u\,du$,

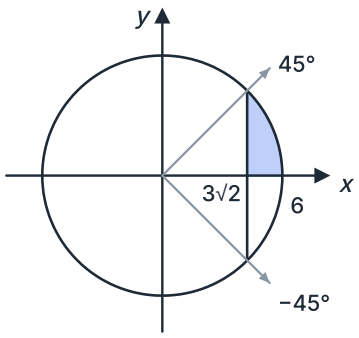
and $\sqrt{36-x^2} = 6\cos u$.

When $x = 3\sqrt{2}$, $u = \frac{\pi}{4}$,

when $x = 6$, $u = \frac{\pi}{2}$.

b The integral is sketched opposite. The shaded area is half the segment subtending an angle of $90°$. Hence

$$\begin{aligned}
\int_{3\sqrt{2}}^{6} \sqrt{36-x^2}\,dx &= \tfrac{1}{2}\times\tfrac{1}{2}\times 6^2\left(\tfrac{\pi}{2}-\sin\tfrac{\pi}{2}\right) \\
&= \tfrac{1}{4}\times 36\times\left(\tfrac{\pi}{2}-1\right) \\
&= 9\left(\tfrac{\pi}{2}-1\right)
\end{aligned}$$



READING WORKED EXAMPLE 21

- In part **a**, $36 - x^2 = 36 - 36\sin^2 u = 36\cos^2 u$, and $\cos u \geq 0$ for $\frac{\pi}{4} \leq u \leq \frac{\pi}{2}$.
- When $x = 3\sqrt{2}$, $\sin u = \frac{3\sqrt{2}}{6} = \frac{1}{\sqrt{2}}$; when $x = 6$, $\sin u = 1$.
- The primitive of $18\cos 2u$ is $\frac{18}{2}\sin 2u = 9\sin 2u$.
- In part **b**, the integral is the area under the semicircle $y = \sqrt{36 - x^2}$ from $x = 3\sqrt{2}$ to $x = 6$. The chord $x = 3\sqrt{2}$ meets the circle at the angles $45°$ and $-45°$, so it subtends $90°$ at the centre.
- The two answers agree, because $9\left(\frac{\pi}{2} - 1\right) = \frac{9\pi}{2} - 9 = \frac{9}{2}(\pi - 2)$.

CONCEPT 3 · choosing the values of u, and other trigonometric substitutions

A NOTE ON THE VALUES OF u

Careful readers may notice a problem here, in that given the value $x = 3\sqrt{2}$, u is determined by $\sin u = \frac{1}{2}\sqrt{2}$, so there are infinitely many possible values of u. A similar problem occurred in the previous worked example, where $0 = 1 - u^2$ had two solutions. These problems arise because the functions involved in the substitutions were $x = 1 - u^2$ and $x = 6\sin u$, whose inverses were not functions. A full account of all this would require substitutions by restrictions of the functions given above so that they had inverse functions. In practice, however, this is rarely necessary, and it is certainly not a concern of this course.

As a rule of thumb, work with positive square roots, and with trigonometric functions, work in the same quadrants as were involved in the definitions of the inverse trigonometric functions in Chapter 17 of the Year 11 book.

10 INTEGRATION USING THE REVERSE CHAIN RULE AND SUBSTITUTION

Here is a summary of Lessons 12D and 12E. The chain rule can be used in reverse in various ways to find integrals.

- The reverse chain rule straightforwardly reverses the steps of differentiation by the chain rule.
- A reverse-chain-rule formula can be developed for each standard form.
- Substitution can be done by, 'Let u be a function of x.'
- Substitution can be done by, 'Let x be a function of u.'
- When substituting into a definite integral, the limits of integration can be substituted as well, so that there is no need ever to return from u to x.

THE RULE OF THUMB IN PRACTICE ($a > 0$)

- $x = 1 - u^2$ and similar: take $u \geq 0$, so that $u = \sqrt{1 - x}$.
- $x = a\sin\theta$: take $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, so $\theta = \sin^{-1}\dfrac{x}{a}$ and $\cos\theta \geq 0$.
- $x = a\cos\theta$: take $0 \leq \theta \leq \pi$, so $\theta = \cos^{-1}\dfrac{x}{a}$ and $\sin\theta \geq 0$.
- $x = a\tan\theta$: take $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, so $\theta = \tan^{-1}\dfrac{x}{a}$ and $\cos\theta > 0$, $\sec\theta > 0$.

WHICH TRIGONOMETRIC SUBSTITUTION? (EXERCISE 12E QUESTIONS 8 TO 12)

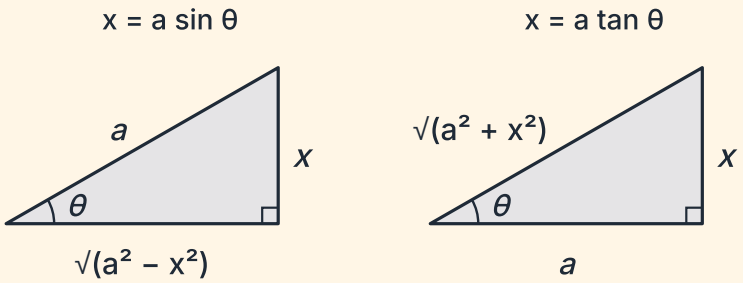
Each substitution turns a sum or difference of squares into a single square, using $\sin^2\theta + \cos^2\theta = 1$ and $\tan^2\theta + 1 = \sec^2\theta$ (Lesson 12C):

$$x = a\sin\theta: \quad a^2 - x^2 = a^2(1 - \sin^2\theta) = a^2\cos^2\theta$$
$$x = a\cos\theta: \quad a^2 - x^2 = a^2(1 - \cos^2\theta) = a^2\sin^2\theta$$
$$x = a\tan\theta: \quad a^2 + x^2 = a^2(1 + \tan^2\theta) = a^2\sec^2\theta$$
$$x = a\sec\theta: \quad x^2 - a^2 = a^2(\sec^2\theta - 1) = a^2\tan^2\theta$$

RETURNING TO x IN AN INDEFINITE INTEGRAL

The answer comes out as a function of θ, such as $\tan\theta$ or $\sin\theta$, and must be written in terms of x. With $x = a\sin\theta$, $\sin\theta = \dfrac{x}{a}$ and $\cos\theta = \sqrt{1 - \sin^2\theta} = \dfrac{\sqrt{a^2 - x^2}}{a}$, so $\tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{x}{\sqrt{a^2 - x^2}}$.

With $x = a\tan\theta$, $\sec\theta = \sqrt{1 + \tan^2\theta} = \dfrac{\sqrt{a^2 + x^2}}{a}$, so $\sin\theta = \dfrac{\tan\theta}{\sec\theta} = \dfrac{x}{\sqrt{a^2 + x^2}}$. The right-angled triangles below give the same results when θ is acute.



TUTOR DEMO **Other substitutions**

a Find $\int \frac{1}{(9+x^2)^{\frac{3}{2}}}\,dx$, using the substitution $x = 3\tan\theta$.

b Evaluate $\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$, using the substitution $x = 2\cos\theta$.

YOUR TURN **Same methods**

a Find $\int \frac{1}{(4-x^2)^{\frac{3}{2}}}\,dx$, using the substitution $x = 2\sin\theta$.

b Evaluate $\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}}\,dx$, using the substitution $x = 2\cos\theta$.

TEXTBOOK HOMEWORK · Exercise 12E

FOUNDATION

1 Consider the integral $I = \int x(x-1)^5 \, dx$, and let $x = u + 1$.

a Show that $dx = du$.

b Show that $I = \int u^5(u+1) \, du$.

c Hence find I.

d Check your answer by differentiating it.

2 Using the same substitution as in the previous question, find:

a $\int \dfrac{x}{\sqrt{x-1}} \, dx$

b $\int \dfrac{x}{(x-1)^2} \, dx$

3 Consider the integral $J = \int x\sqrt{x+1}\,dx$, and let $x = u^2 - 1$.

a Show that $dx = 2u\,du$.

b Show that $J = 2\int (u^4 - u^2)\,du$.

c Hence find J.

d Check your answer by differentiating it.

4 Using the same substitution as in the previous question, find:

a $\int x^2\sqrt{x+1}\,dx$

b $\int \dfrac{2x+3}{\sqrt{x+1}}\,dx$

5 Find each indefinite integral using the given substitution.

a $\displaystyle\int \frac{x-2}{x+2}\,dx$ (Let $x = u - 2$.)

b $\displaystyle\int \frac{2x+1}{\sqrt{2x-1}}\,dx$ (Let $x = \frac{1}{2}(u+1)$.)

c $\displaystyle\int 3x\sqrt{4x-5}\,dx$ (Let $x = \frac{1}{4}(u^2+5)$.)

d $\displaystyle\int \frac{1}{1+\sqrt{x}}\,dx$ (Let $x = (u-1)^2$.)

6 Evaluate, using the given substitution:

a $\displaystyle\int_0^1 x(x+1)^3\,dx$ (Let $x = u - 1$.)

b $\displaystyle\int_0^{\frac{1}{2}} \frac{1+x}{1-x}\,dx$ (Let $x = 1 - u$.)

c $\displaystyle\int_0^1 \frac{3x}{\sqrt{3x+1}}\,dx$ (Let $x = \frac{1}{3}(u-1)$.)

d $\displaystyle\int_0^1 \frac{2-x}{(2+x)^3}\,dx$ (Let $x = u - 2$.)

e $\displaystyle\int_0^4 x\sqrt{4-x}\,dx$ (Let $x = 4 - u^2$.)

f $\displaystyle\int_1^5 \frac{x}{(2x-1)^{\frac{3}{2}}}\,dx$ (Let $x = \frac{1}{2}(u^2+1)$.)

g $\displaystyle\int_0^4 \frac{1}{3+\sqrt{x}}\,dx$ (Let $x = (u-3)^2$.)

h $\displaystyle\int_0^7 \frac{x^2}{\sqrt[3]{x+1}}\,dx$ (Let $x = u^3 - 1$.)

DEVELOPMENT

7 **a** Consider the integral $I = \int \frac{1}{\sqrt{5-4x-x^2}}\,dx$, and let $x = u - 2$.

Show that $I = \int \frac{1}{\sqrt{9-u^2}}\,du$, and hence find I.

b Use a similar approach to find:

(i) $\int \frac{1}{x^2+2x+4}\,dx$ (Let $x = u - 1$.)

(ii) $\int \frac{1}{\sqrt{4-2x-x^2}}\,dx$ (Let $x = u - 1$.)

(iii) $\int_1^2 \frac{1}{\sqrt{3+2x-x^2}}\,dx$ (Let $x = u + 1$.)

(iv) $\int_3^7 \frac{1}{x^2-6x+25}\,dx$ (Let $x = u + 3$.)

8 **a** Consider the integral $J = \int \dfrac{1}{\sqrt{4-x^2}}\,dx$, and let $x = 2\sin\theta$.

Show that $J = \int 1\,d\theta$, and hence show that $J = \sin^{-1}\frac{x}{2} + C$.

b Using a similar approach, find:

(i) $\int \dfrac{1}{9+x^2}\,dx$ (Let $x = 3\tan\theta$.)

(ii) $\int \dfrac{-1}{\sqrt{3-x^2}}\,dx$ (Let $x = \sqrt{3}\cos\theta$.)

(iii) $\int \dfrac{1}{\sqrt{1-4x^2}}\,dx$ (Let $x = \frac{1}{2}\sin\theta$.)

(iv) $\int \dfrac{1}{1+16x^2}\,dx$ (Let $x = \frac{1}{4}\tan\theta$.)

(v) $\int_0^3 \dfrac{1}{\sqrt{36-x^2}}\,dx$ (Let $x = 6\sin\theta$.)

(vi) $\int_0^{\frac{2}{3}} \dfrac{1}{4+9x^2}\,dx$ (Let $x = \frac{2}{3}\tan\theta$.)

9 **a** Consider the integral $I = \int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,dx$, and let $x = \sin\theta$.

Show that $I = \int \sec^2\theta\,d\theta$, and hence show that $I = \frac{x}{\sqrt{1-x^2}} + C$.

b Similarly, use the given substitution to find:

(i) $\int \frac{1}{(4+x^2)^{\frac{3}{2}}}\,dx$ (Let $x = 2\tan\theta$.)

(ii) $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx$ (Let $x = \sin\theta$.)

(iii) $\int_0^2 \sqrt{4-x^2}\,dx$ (Let $x = 2\sin\theta$.)

(iv) $\int \frac{1}{x^2\sqrt{25-x^2}}\,dx$ (Let $x = 5\cos\theta$.)

(v) $\int \frac{1}{x^2\sqrt{9+x^2}}\,dx$ (Let $x = 3\tan\theta$.)

(vi) $\int_2^4 \frac{1}{x^2\sqrt{x^2-4}}\,dx$ (Let $x = 2\sec\theta$.)

10 Find the equation of the curve $y = f(x)$ if $f'(x) = \dfrac{\sqrt{x^2 - 9}}{x}$ and $f(3) = 0$.

(Hint: Use the substitution $x = 3\sec\theta$.)

11 Find the exact area of the region bounded by $y = \dfrac{x^3}{\sqrt{3 - x^2}}$, the x-axis and the line $x = 1$.

(Hint: Use the substitution $x = \sqrt{3}\sin\theta$, followed by the substitution $u = \cos\theta$.)

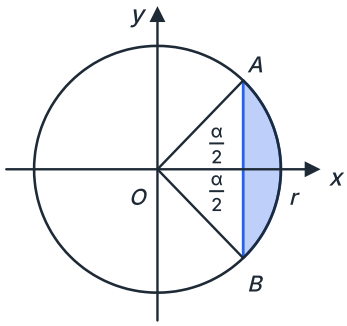
12 [These are confirmations rather than proofs, because the calculus of trigonometric functions was developed on the basis of the formulae in parts **a** and **b**.]

a Use integration to confirm that the area of a circle is πr^2.

(Hint: Find the area bounded by the semi-circle $y = \sqrt{r^2 - x^2}$ and the x-axis and double it. Use the substitution $x = r\sin\theta$.)

b

The shaded area in the diagram to the right is the segment of a circle of radius r cut off by the chord AB subtending an angle α at the centre O.



(i) Show that the area is $I = 2\displaystyle\int_{r\cos\frac{1}{2}\alpha}^{r} \sqrt{r^2 - x^2}\,dx$.

(ii) Let $x = r\cos\theta$, and show that $I = -2r^2\displaystyle\int_{\frac{1}{2}\alpha}^{0} \sin^2\theta\,d\theta$.

(iii) Hence confirm that $I = \frac{1}{2}r^2(\alpha - \sin\alpha)$.

c Use a similar approach to confirm that the area of the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is πab. Then justify the formula by regarding the ellipse as the unit circle stretched horizontally by a factor of a and vertically by a factor of b.

ENRICHMENT

13 a Use the substitution $x = -u$ to show that $\displaystyle\int_{-2}^{2} \frac{x^2}{e^x + 1}\,dx = \int_{-2}^{2} \frac{x^2 e^x}{e^x + 1}\,dx$.

b Hence find $\displaystyle\int_{-2}^{2} \frac{x^2}{e^x + 1}\,dx$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **18 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q5** **1 mark**

Which of the following is equal to $\int_0^{\sqrt{3}} \frac{1}{(x^2+1)^2}\,dx$ after applying the substitution $x = \tan\theta$?

(A) $\int_0^{\frac{\pi}{3}} \cos^2\theta\,d\theta$

(B) $\int_0^{\frac{\pi}{3}} \sec^2\theta\,d\theta$

(C) $\int_0^{\frac{\pi}{6}} \cos^2\theta\,d\theta$

(D) $\int_0^{\frac{\pi}{6}} \sec^2\theta\,d\theta$

Hint: what does each part of the integral look like once x is replaced?

PP 2 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q4** **1 mark**

What is the value of $\int_0^{\frac{\sqrt{3}}{2}} \sqrt{1-y^2}\,dy$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6} + \frac{\sqrt{3}}{8}$

(D) $\frac{\pi}{3} + \frac{\sqrt{3}}{8}$

Hint: what does the integrand look like once y is replaced?

PP 3 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d)** 4 marks

Let $f(x) = \dfrac{2x}{\sqrt{1-x^2}}$

(i) For what values of x is $f(x)$ undefined? 1

(ii) Find $\displaystyle\int_0^{\frac{1}{2}} \frac{2x\,dx}{\sqrt{1-x^2}}$ using the substitution $x = \sin u$. 3

Hint: for which values of x does the square root in the denominator cause trouble?

PP 4 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q12(b)** 3 marks

Find the following definite integral using the substitution $x = \cos\theta$.
Express your answer in exact value.

$$\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}}\,dx$$

Hint: what does the integral look like once x is replaced?

PP 5 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(b)** 4 marks

(i) Using the substitution $x = \cos 2\theta$, show $\displaystyle\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx = \int_0^{\frac{\pi}{4}} 4\sin^2\theta\,d\theta$ 2

(ii) Hence evaluate in simplest exact form $\displaystyle\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$ 2

Hint: what does the fraction under the square root look like once x is replaced?

PP 6 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(a)** 5 marks

Using the substitution $x = (u-4)^2 + 1$, or otherwise, evaluate $\displaystyle\int_2^5 \frac{1}{4+\sqrt{x-1}}\,dx$

Hint: what does the denominator look like once x is replaced?

LESSON 12F

Volumes of rotation

When a region in the coordinate plane is rotated about the x-axis or the y-axis, it generates a solid. Slicing the solid into thin circular slices gives a simple integral for its volume. Rotating the region between two curves needs a subtraction of volumes, and the same method finally proves the formulae for the volumes of cones and spheres.

CONCEPT 1 · rotating a region about the x-axis

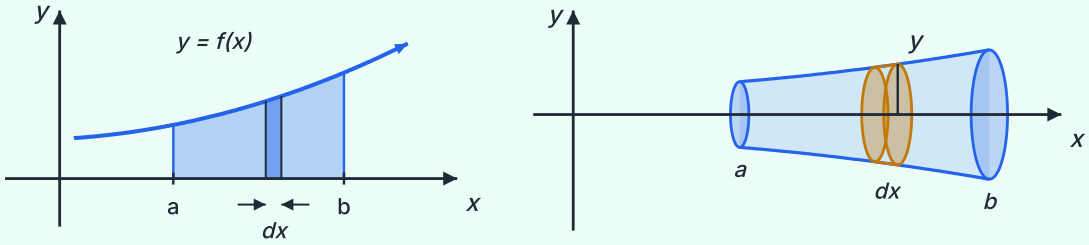
SOLIDS OF REVOLUTION

When a region in the coordinate plane is rotated about either the x-axis or the y-axis, a solid region in three dimensions is generated, called a *solid of revolution*. The process is similar to shaping wood on a lathe, or making pottery on a wheel, because such shapes have rotational symmetry and circular cross-sections.

The volumes of such solids can be found using a simple integration formula. The well-known formulae for the volumes of cones and spheres can finally be proven by this method.

SLICING THE SOLID

The first diagram below shows the region under the curve $y = f(x)$ in the interval $a \leq x \leq b$, and the second shows the solid generated when this region is rotated about the x-axis.



Imagine the solid sliced like salami perpendicular to the x-axis into infinitely many circular slices, each of width dx. One of the slices is shown in the right-hand diagram. The vertical strip in the left-hand diagram is what generates this slice when it is rotated about the x-axis.

Now	radius of circular slice $= y$,	the height of the strip,
so	area of circular slice $= \pi y^2$,	because it is a circle.
	The slice is a thin cylinder of infinitesimal thickness dx,	
so	volume of circular slice $= \pi y^2\, dx$	(area × thickness).

To get the total volume, we add all the slices from $x = a$ to $x = b$, so

$$\text{volume of solid} = \int_a^b \pi y^2\, dx.$$

11 VOLUMES OF REVOLUTION ABOUT THE x-AXIS

The volume of the solid generated when the region between a curve and the x-axis from $x = a$ to $x = b$ is rotated about the x-axis is

$$\text{volume} = \int_a^b \pi y^2 \, dx \text{ cubic units.}$$

TWO REMARKS

- If the curve is below the x-axis, so that y is negative, then the volume calculated is still positive, because y^2 rather than y occurs in the formula.
- Unless other units are specified, 'cubic units' (u^3) should be used, by analogy with the areas of regions discussed in Section 3F.

THE STEPS

(1) Square the equation of the curve, so that y^2 is written in terms of x. (2) Write $V = \pi \int_a^b y^2 \, dx$, with the limits on the x-axis. (3) Find the primitive. (4) Substitute the upper and lower limits and subtract. (5) Write the units.

TUTOR DEMO **A region cut off a semicircle**

The region cut off the semicircle $y = \sqrt{25 - x^2}$ by the line $x = 3$ (the part to the right of the line) is rotated about the x-axis. Find the volume generated.

YOUR TURN **Same method**

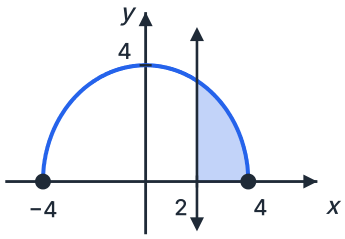
The region cut off the semicircle $y = \sqrt{9 - x^2}$ by the line $x = 1$ (the part to the right of the line) is rotated about the x-axis. Find the volume generated.

WORKED EXAMPLE 22 **Keep for homework**

The shaded region cut off the semicircle $y = \sqrt{16 - x^2}$ by the line $x = 2$ is rotated about the x-axis. Find the volume generated.

SOLUTION

Squaring, $y^2 = 16 - x^2$.



$$\begin{aligned}
\text{so} \quad \text{volume} &= \int_2^4 \pi y^2 \, dx \\
&= \pi \int_2^4 (16 - x^2) \, dx \\
&= \pi \left[16x - \tfrac{1}{3}x^3\right]_2^4 \\
&= \pi \left(\left(16 \times 4 - \tfrac{1}{3} \times 4^3\right) - \left(16 \times 2 - \tfrac{1}{3} \times 2^3\right) \right) \\
&= \pi \left(64 - \tfrac{64}{3} - 32 + \tfrac{8}{3}\right) \\
&= \pi \left(32 - \tfrac{56}{3}\right) \\
&= \tfrac{40\pi}{3} \text{ cubic units.}
\end{aligned}$$

LIMITS AND ARITHMETIC

The region runs from the line $x = 2$ to $x = 4$, where the semicircle meets the x-axis, so these are the limits. In the last step, $32 - \frac{56}{3} = \frac{96}{3} - \frac{56}{3} = \frac{40}{3}$.

CONCEPT 2 · volumes of revolution about the y-axis

EXCHANGE x AND y

To calculate the volume when a region is rotated about the y-axis, we exchange x and y.

12 VOLUMES OF REVOLUTION ABOUT THE y-AXIS

The volume of the solid generated when the region between a curve and the y-axis from $y = a$ to $y = b$ is rotated about the y-axis is

$$\text{volume} = \int_a^b \pi x^2 \, dy \text{ cubic units.}$$

MAKE x^2 THE SUBJECT

When y is given as a function of x, the equation will need to be written with x^2 as the subject.

HORIZONTAL SLICES

The slices are now horizontal circular slices of radius x and thickness dy, so the limits are values of y. For example, $y = 3x^2$ gives $x^2 = \frac{1}{3}y$.

TUTOR DEMO **Rotation about the y-axis**

Find the volume of the solid formed by rotating the region between $y = 2x^2$ and the line $y = 8$ about the y-axis.

YOUR TURN **Same method**

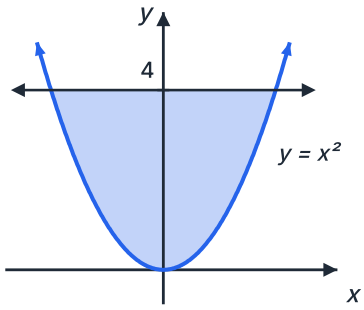
Find the volume of the solid formed by rotating the region between $y = x^2$ and the line $y = 9$ about the y-axis.

WORKED EXAMPLE 23 **Keep for homework**

Find the volume of the solid formed by rotating the region between $y = x^2$ and the line $y = 4$ about the y-axis.

SOLUTION

Because $x^2 = y$,



$$\begin{aligned}\text{volume} &= \int_0^4 \pi x^2\, dy \\ &= \pi \int_0^4 y\, dy \\ &= \pi \left[\tfrac{1}{2}y^2\right]_0^4 \\ &= \pi\left(\tfrac{1}{2} \times 4^2 - \tfrac{1}{2} \times 0^2\right) \\ &= 8\pi \text{ cubic units.}\end{aligned}$$

LIMITS

The limits are $y = 0$ (the vertex of the parabola) and $y = 4$ (the line).

CONCEPT 3 · volume by subtraction

THE REGION BETWEEN TWO CURVES

When rotating the region between two curves lying above the x-axis, the two integrals need to be subtracted, as if the outer volume has been formed first, and the inner volume then cut away from it. The two volumes can always be calculated separately and subtracted, but if the two integrals have the same limits of integration, it may be more convenient to combine them.

13 ROTATING THE REGION BETWEEN CURVES

- The volume of the solid generated when the region between two curves from $x = a$ to $x = b$ is rotated about the x-axis is

$$\text{volume} = \int_a^b \pi\left(y_2^2 - y_1^2\right) dx \qquad (\text{where } y_2 > y_1 > 0).$$

- Similarly, the volume of the solid generated when the region between two curves from $y = a$ to $y = b$ is rotated about the y-axis is

$$\text{volume} = \int_a^b \pi\left(x_2^2 - x_1^2\right) dy \qquad (\text{where } x_2 > x_1 > 0).$$

SQUARE EACH CURVE FIRST, THEN SUBTRACT

Each slice is now a washer: a disc of radius y_2 with a hole of radius y_1. Its area is $\pi y_2^2 - \pi y_1^2$, which is **not** $\pi(y_2 - y_1)^2$.

NOTE

One would not normally expect the volumes of revolution about the two different axes to be equal. This is because an element of area will generate a larger element of volume if it is moved further away from the axis of rotation.

TUTOR DEMO **Both axes**

The curve $y = 9 - x^2$ meets the y-axis at $A(0, 9)$ and the x-axis at $B(3, 0)$. Find the volume generated when the region between the curve and the line AB is rotated:

a about the x-axis, b about the y-axis.

YOUR TURN **Same method**

The curve $y = 1 - x^2$ meets the y-axis at $A(0, 1)$ and the x-axis at $B(1, 0)$. Find the volume generated when the region between the curve and the line AB is rotated:

a about the x-axis,

b about the y-axis.

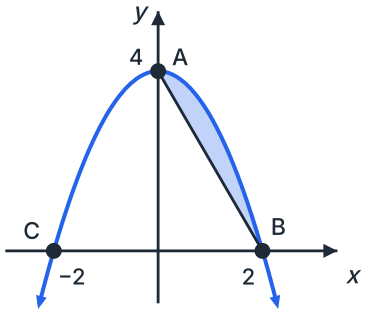
WORKED EXAMPLE 24 **Keep for homework**

The curve $y = 4 - x^2$ meets the y-axis at $A(0, 4)$ and the x-axis at $B(2, 0)$ and $C(-2, 0)$. Find the volume generated when the region between the curve and the line AB is rotated:

a about the x-axis, b about the y-axis.

SOLUTION

a The parabola is $y_2 = 4 - x^2$, and the line AB is $y_1 = 4 - 2x$,



THE LINE AND THE OUTER CURVE

The line AB has gradient $\dfrac{0-4}{2-0} = -2$ and y-intercept 4. For $0 \le x \le 2$, $y_2 - y_1 = 2x - x^2 = x(2-x) \ge 0$, so the parabola is the outer curve.

$$\begin{aligned}
\text{so} \quad \text{volume} &= \int_0^2 \pi\left(y_2^2 - y_1^2\right) dx \\
&= \pi \int_0^2 \left(\left(16 - 8x^2 + x^4\right) - \left(16 - 16x + 4x^2\right)\right) dx \\
&= \pi \int_0^2 \left(x^4 - 12x^2 + 16x\right) dx \\
&= \pi \left[\tfrac{1}{5}x^5 - 4x^3 + 8x^2\right]_0^2 \\
&= \pi\left(\left(\tfrac{1}{5} \times 2^5 - 4 \times 2^3 + 8 \times 2^2\right) - 0\right) \\
&= \pi\left(\tfrac{32}{5} - 32 + 32\right) \\
&= \tfrac{32\pi}{5} \text{ cubic units.}
\end{aligned}$$

SQUARING BOTH

The second line uses

$y_2^2 = \left(4 - x^2\right)^2 = 16 - 8x^2 + x^4$,

$y_1^2 = (4 - 2x)^2 = 16 - 16x + 4x^2$.

b The parabola is $x_2^2 = 4 - y$, and the line is $x_1 = 2 - \frac{1}{2}y$,

$$\begin{aligned}
\text{so} \quad \text{volume} &= \int_0^4 \pi\left(x_2^2 - x_1^2\right) dy \\
&= \pi \int_0^4 \left((4 - y) - \left(4 - 2y + \tfrac{1}{4}y^2\right)\right) dy \\
&= \pi \int_0^4 \left(-\tfrac{1}{4}y^2 + y\right) dy \\
&= \pi \left[-\tfrac{1}{12}y^3 + \tfrac{1}{2}y^2\right]_0^4 \\
&= \pi\left(\left(-\tfrac{1}{12} \times 4^3 + \tfrac{1}{2} \times 4^2\right) - 0\right) \\
&= \pi\left(-\tfrac{16}{3} + 8\right) \\
&= \tfrac{8\pi}{3} \text{ cubic units.}
\end{aligned}$$

MAKING x THE SUBJECT

From $y = 4 - x^2$, $x^2 = 4 - y$. From $y = 4 - 2x$, $2x = 4 - y$, so $x = 2 - \frac{1}{2}y$, and $x_1^2 = \left(2 - \frac{1}{2}y\right)^2 = 4 - 2y + \frac{1}{4}y^2$. The limits are $y = 0$ (at B) and $y = 4$ (at A).

COMPARE

The two volumes are different, as the note above says they usually will be.

CONCEPT 4 · cones and spheres

PROVING THE FORMULAE

The formulae for the volumes of cones and spheres may have been learnt earlier, but they cannot be proven without arguments involving integration. The proofs of both results are developed in Question 22 of the exercise, and these questions should be carefully worked through.

HOW THE SOLIDS ARE GENERATED

- A right circular cone of height h and radius r is generated by rotating the region under the line $y = \dfrac{rx}{h}$ from $x = 0$ to $x = h$ about the x-axis.
- A sphere of radius r is generated by rotating the region under the semicircle $y = \sqrt{r^2 - x^2}$ from $x = -r$ to $x = r$ about the x-axis.

14 VOLUME OF CONES AND SPHERES

- For a cone, $V = \frac{1}{3}\pi r^2 h$.
- For a sphere, $V = \frac{4}{3}\pi r^3$.

TUTOR DEMO **A cone by integration**

The region under the line $y = \frac{1}{2}x$ from $x = 0$ to $x = 6$ is rotated about the x-axis to form a cone. Find the radius and height of the cone, find its volume by integration, and check the answer with Box 14.

YOUR TURN **A sphere by integration**

The region under the semicircle $y = \sqrt{16 - x^2}$ from $x = -4$ to $x = 4$ is rotated about the x-axis to form a sphere. Find its radius, find its volume by integration, and check the answer with Box 14.

THE GENERAL PROOFS

There is no textbook example for Box 14. The general proofs, with r and h in place of numbers, are Question 22 of the homework.

TEXTBOOK HOMEWORK · Exercise 12F

FOUNDATION

1 a Sketch the region bounded by the line $y = 3x$ and the x-axis between $x = 0$ and $x = 3$.

b When this region is rotated about the x-axis, a right circular cone is formed. Find the radius and height of the cone and hence find its volume.

c Evaluate $\pi \int_0^3 y^2\,dx = \pi \int_0^3 9x^2\,dx$ in order to check your answer.

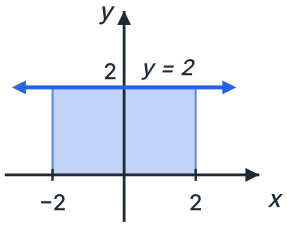
2 a Sketch the region bounded by the curve $y = \sqrt{9 - x^2}$ and the x-axis between $x = -3$ and $x = 3$.

b When this region is rotated about the x-axis, a sphere is formed. Find the radius of the sphere and hence find its volume.

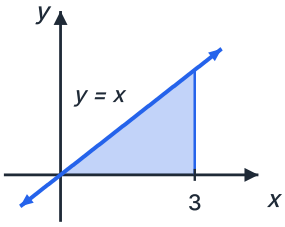
c Evaluate $\pi \int_{-3}^3 y^2\,dx = \pi \int_{-3}^3 \left(9 - x^2\right)\,dx$ in order to check your answer.

3 Calculate the volume generated when each shaded region is rotated about the x-axis.

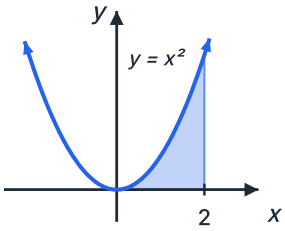
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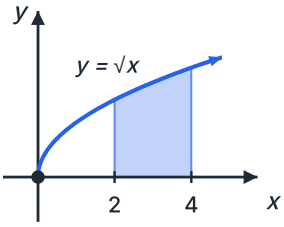
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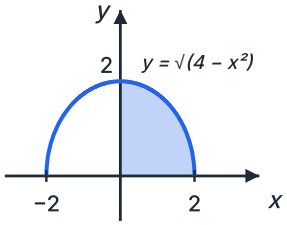
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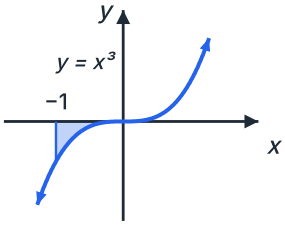
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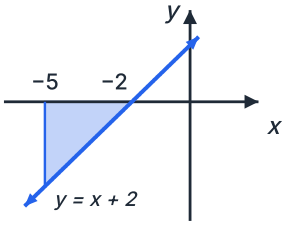
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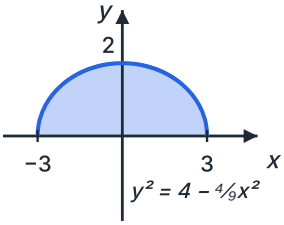
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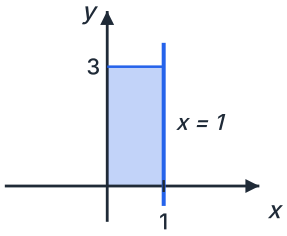


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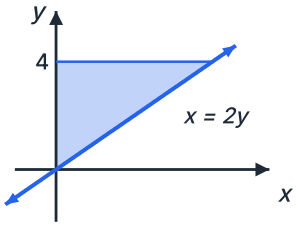


4 Calculate the volume generated when each shaded region is rotated about the y-axis.

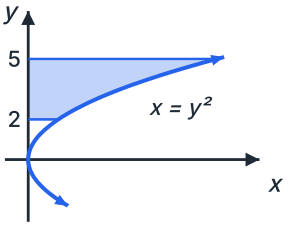
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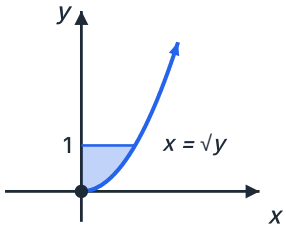
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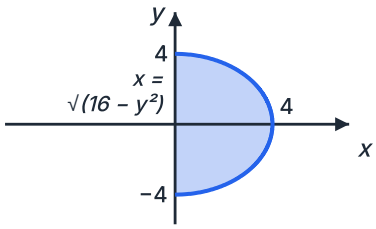
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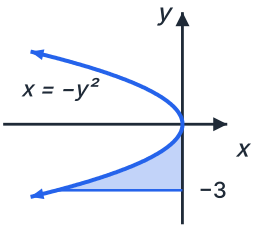
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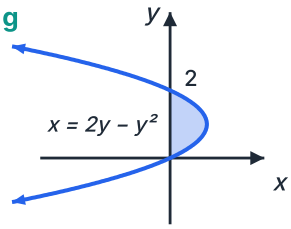
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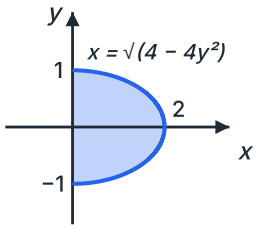
f



g



h



DEVELOPMENT

5 The region between the curve $y = e^x$ and the x-axis from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume of the solid generated.

6 Find the volume generated when the region between the curve $y = \dfrac{1}{\sqrt{x}}$ and the x-axis, from $x = 2$ to $x = 4$, is rotated about the x-axis.

7 The region between the curve $y = \dfrac{1}{x^2}$ (called a *truncus*) and the y-axis from $y = 1$ to $y = 6$ is rotated about the y-axis. Calculate the exact volume of the solid formed.

8 a Write $\tan^2 x$ in terms of $\sec^2 x$.

b The region bounded by the curve $y = \tan x$, the x-axis and the vertical line $x = \frac{\pi}{3}$ is rotated about the x-axis. Find the volume of the solid generated.

9 a Write $\sin^2 x$ in terms of $\cos 2x$.

b The region bounded by the curve $y = \sin x$, the x-axis and the vertical line $x = \frac{\pi}{2}$ is rotated about the x-axis. Find the volume of the solid generated.

10 Find the volume of the solid formed by rotating the region with the given boundaries about the x-axis. (A sketch of each region will be needed.)

a $y = x + 3,\ x = 3,\ x = 5$ and $y = 0$

b $y = 1 + \sqrt{x},\ x = 1,\ x = 4$ and $y = 0$

c $y = 5x - x^2$ and $y = 0$

d $y = x^3 - x$ and $y = 0$

11 Find the volume of the solid formed by rotating the region with the given boundaries about the y-axis. (A sketch of each region will be needed.)

a $x = y - 2,\ y = 3,\ x = 1$ and $x = 0$

b $x = y^2 + 1,\ y = 0,\ y = 1$ and $x = 0$

c $x = y(y - 3)$ and $x = 0$

d $y = 1 - x^2$ and $y = 0$

12 A vat is designed by rotating about the x-axis the region between the curve $y = 1 + \dfrac{1}{x}$ and the x-axis from $x = \frac{1}{2}$ to $x = 3$. Show that its volume is $\dfrac{\pi}{6}(25 + 12\ln 6)$ u^3.

13 A metal stud is created by rotating about the x-axis the region contained between the curve $y = e^x - e^{-x}$ and the x-axis from $x = 0$ to $x = \frac{1}{2}$. Show that the volume of the stud is $\frac{\pi}{2}\left(e - 2 - e^{-1}\right)$ u^3.

14 A champagne flute is designed by rotating about the x-axis the region between the curve $y = 4 + 4\sin\frac{x}{4}$ and the x-axis from $x = 4\pi$ to $x = 6\pi$. Find, correct to 4 significant figures, the capacity of the flute, if 1 unit = 1 cm.

15 The region R is bounded by the parabola $y = x^2$ and the x-axis from $x = 0$ to $x = 4$.

a Find the volume of the cylinder formed when the region between the line $x = 4$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

b Find the volume of the solid formed when the region between the parabola $y = x^2$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

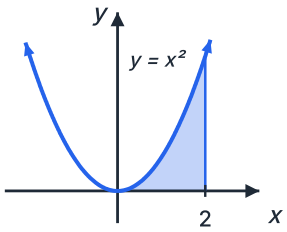
c Hence find the volume of the solid formed when R is rotated about the y-axis.

16 Find the volume of the solid generated by rotating each region about:

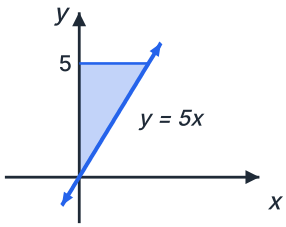
i the x-axis,

ii the y-axis. (Hint: In some cases a subtraction of volumes will be necessary.)

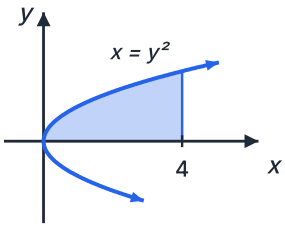
a



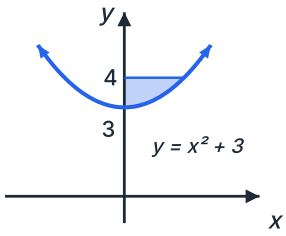
b



c



d



17 **a** Sketch the region bounded by the curves $y = x^2$ and $y = x^3$.

b Find the volume of the solid generated when this region is rotated about:
(i) the x-axis, (ii) the y-axis.

18 **a** On the same number plane sketch the graphs of $xy = 5$ and $x + y = 6$, clearly indicating their points of intersection.

b Find the volume of the solid generated when the region bounded by the two curves is rotated about the x-axis.

19 The region bounded by the hyperbola $y = 2 - \dfrac{2}{x}$, the vertical line $x = 1$ and the horizontal line $y = 1$ is rotated about the x-axis. Find the volume of the solid formed.

20 **a** Find the equation of the tangent to the curve $y = x^3 + 2$ at the point where $x = 1$.

b Draw a diagram showing the region bounded by the curve, the tangent and the y-axis.

c Calculate the volume of the solid formed when this region is rotated about:
(i) the x-axis, (ii) the y-axis.

21 A rubber washer is generated by rotating the region between the curve $y = \log_e x$, the x-axis and the line $x = 2$ about the y-axis. Find the exact volume of the washer.

22 In this question some standard volume formulae will be proven.

a A right circular cone of height h and radius r is generated by rotating about the x-axis the region bounded by the line $y = \dfrac{rx}{h}$, the x-axis and the line $x = h$. Show that the volume of the cone is $\frac{1}{3}\pi r^2 h$.

b A cylinder of height h and radius r is generated by rotating about the x-axis the region bounded by the line $y = r$, the x-axis, the y-axis and the line $x = h$. Show that the volume of the cylinder is $\pi r^2 h$.

c (i) A sphere of radius r is generated by rotating about the x-axis the region between the semi-circle $y = \sqrt{r^2 - x^2}$ and the x-axis. Show that the volume of the sphere is $\frac{4}{3}\pi r^3$.

(ii) A spherical cap of height h is formed by rotating about the x-axis the region between the semi-circle $y = \sqrt{r^2 - x^2}$ and the x-axis from $x = r - h$ to $x = r$. Show that the volume of the cap is $\frac{1}{3}\pi h^2(3r - h)$.

ENRICHMENT

23

a Multiply $\sec\theta$ by $\dfrac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta}$, and hence find $\displaystyle\int \sec\theta\, d\theta$.

b The region R is bounded by $y=\dfrac{x}{\sqrt{x^2+16}}$, the x-axis and the line $x=4$. Show that the volume generated by rotating R about the y-axis is $16\pi\left(\sqrt{2}-\ln\left(\sqrt{2}+1\right)\right)$ units^3. (Hint: Use the substitution $y=\sin\theta$ and the result in part **a**.)

24 Consider the curves $f(x) = x^n$ and $f(x) = x^{n+1}$, where n is a positive integer.

a Find the points of intersection of the curves.

b Show that the volume V_n of the solid generated when the region bounded by the two curves is rotated around the x-axis is given by $\pi\left(\dfrac{1}{2n+1} - \dfrac{1}{2n+3}\right)$ cubic units.

c Describe the solid whose volume is given by $\lim\limits_{n\to\infty}\left(V_1 + V_2 + V_3 + \cdots + V_n\right)$.

d Find the volume of the solid in part **c**.

e Deduce that the series $\dfrac{1}{3\times 5} + \dfrac{1}{5\times 7} + \dfrac{1}{7\times 9} + \cdots$ has a limiting sum of $\dfrac{1}{6}$.

25 **a** Sketch the region bounded by the parabola $y = -x^2 + 6x - 8$ and the x-axis.

b By completing the square, or otherwise, show that the equation of the curve is $x = 3 + \sqrt{1 - y}$ for $3 \leq x \leq 4$, and $x = 3 - \sqrt{1 - y}$ for $2 \leq x \leq 3$.

c The region in part **a** is rotated about the y-axis. Show that the volume of the solid formed is given by $V = \int_0^1 12\pi\sqrt{1 - y}\, dy$, and hence calculate the exact volume.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q8** **1 mark**

The region bounded by the curve $y = e^{\frac{x}{2}}$ and the y-axis for $1 \leq y \leq 2$ is rotated about the y-axis to form a solid.

Which of the following integrals would give the volume of the solid?

A. $V = 4\pi \int_1^2 y \ln y \, dy$

B. $V = 4\pi \int_1^2 \ln y^2 \, dy$

C. $V = 4\pi \int_1^2 xe^x \, dx$

D. $V = 4\pi \int_1^2 (\ln y)^2 \, dy$

Hint: which axis is the solid rotated about?

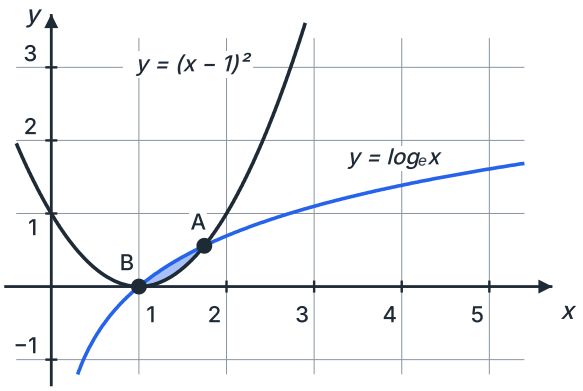
PP 2 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q12(a)** **2 marks**

The region enclosed by the curve $y = \dfrac{2}{\sqrt{x-7}}$ and the x-axis between $x = 8$ and $x = 10$ is rotated about the x-axis. Find the exact value of the volume of the solid formed.

Hint: what region is being rotated?

PP 3 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q8** 1 mark

The graph below shows the graph of $y = (x-1)^2$ and $y = \log_e x$. They intersect at the point $B(1, 0)$ and $A(a, \log_e a)$. The shaded area between these two curves is rotated about the y axis. Which integral represents the volume of the solid of revolution formed?



A. $V = \pi \int_0^{\ln a} \left(\sqrt{y} + 1 - e^y\right)^2 dy$

B. $V = \pi \int_0^{\ln a} \left(y + 2\sqrt{y} + 1 - e^{2y}\right) dy$

C. $V = \pi \int_1^{a} (\ln x)^2 - (x-1)^4 \, dx$

D. $V = \pi \int_0^{\ln a} \left(\sqrt{y} + 1 - e^y\right) dy$

Hint: what does the diagram show?

PP 4 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(a)** 3 marks

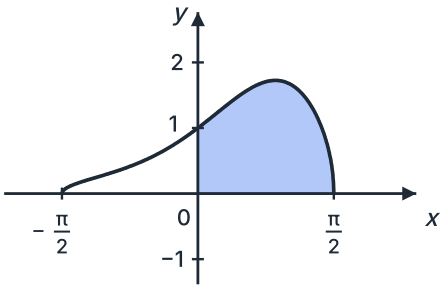
The region between the graph of $f(x) = \sin x \cos x$ and the x – axis over the interval $x \in \left[0, \frac{\pi}{2}\right]$ is revolved about the x – axis. Find the volume of the resulting solid. You may sketch a diagram first.

Hint: what is being integrated here?

PP 5 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q14(b)** 4 marks

(i) Show that $\dfrac{d}{dx}\left(e^{2\sin x}\right) = 2\cos x\, e^{2\sin x}$ 1

(ii) The region bounded by the graph $y = \sqrt{\cos x}\, e^{\sin x}$ and the x-axis between $x = 0$ and $x = \dfrac{\pi}{2}$ is rotated about the x-axis to form a solid. 3



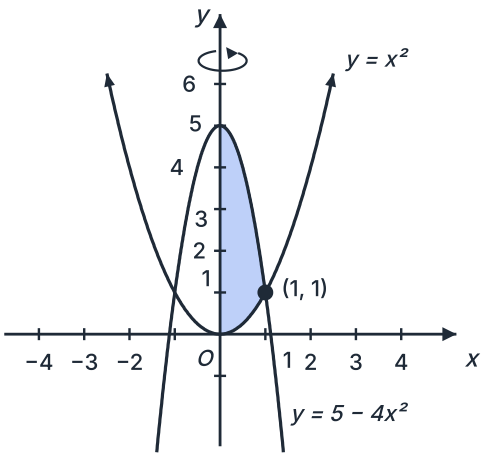
Using the result from part i), find the exact volume of this solid.

Hint: what does part (i) give you?

PP 6 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q13(c)** 3 marks

A solid of revolution is to be formed by rotating the shaded area shown between the graphs $y = x^2$ and $y = 5 - 4x^2$ about the **y-axis**.

Find the exact volume of the solid formed.



Hint: what does the diagram show?

PP 7 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(e)** 3 marks

Find the volume of the solid of revolution formed when the region in the first quadrant bounded by the y-axis, $y = \frac{1}{2}x + 3$, and $y = x^2$ is rotated about the y-axis.

Hint: what does the region look like?

PP 8 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q12(a)** 4 marks

Consider the parabola $x^2 = 8(y-3)$.

(i) Sketch the parabola labelling its vertex. 1

(ii) The area bounded by the parabola and the line $y = 5$ is rotated about the x-axis. Find the volume of the solid formed. 3

Hint: what region is being rotated?

CHAPTER 12 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 12A to 12F.

1 Find the derivative of:

a $y = \sin^{-1} 3x$

b $y = \tan^{-1} \dfrac{x}{3}$

c $y = \cos^{-1}(1 - x)$

d $y = x^2 \tan^{-1} x$

e $y = \tan^{-1}\left(\frac{1}{2}x + 1\right)$

f $y = \sin^{-1} \dfrac{1}{x}$

2 Show that $y = \tan^{-1} x$ satisfies the differential equation $(1 + x^2)y'' + 2xy' = 0$.

4 Find:

a $\displaystyle\int \frac{1}{3 + x^2}\, dx$

b $\displaystyle\int \frac{1}{\sqrt{3 - x^2}}\, dx$

c $\displaystyle\int \frac{1}{9 + 4x^2}\, dx$

d $\displaystyle\int \frac{-1}{\sqrt{16 - 9x^2}}\, dx$

5 Evaluate:

a $\displaystyle\int_{\frac{1}{3}}^{\frac{1}{\sqrt{3}}} \frac{1}{1+9x^2}\,dx$

b $\displaystyle\int_{-\frac{3}{4}}^{\frac{3}{4}} \frac{1}{\sqrt{3-4x^2}}\,dx$

6 Find:

a $\displaystyle\int \cos^2 x\,dx$

b $\displaystyle\int \sin^2 x\,dx$

c $\displaystyle\int \cos^2 2x\,dx$

d $\displaystyle\int \sin^2 4x\,dx$

7 Find the exact value of:

a $\displaystyle\int_0^{\frac{\pi}{3}} \sin^2 3x\,dx$

b $\displaystyle\int_0^{\frac{\pi}{6}} \cos^2 \tfrac{1}{2}x\,dx$

8 Show that $\displaystyle\int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\frac{\pi}{2}} \cos^2 2x\,dx = \int_0^{\frac{\pi}{2}} \cos^2 4x\,dx = \frac{\pi}{4}$.

9 Find each indefinite integral using the given substitution.

a $\displaystyle\int 5(5x-1)^5\,dx$ [Let $u = 5x - 1$.]

b $\displaystyle\int 2x\left(x^2+2\right)^2\,dx$ [Let $u = x^2 + 2$.]

c $\displaystyle\int \frac{4x^3}{\left(x^4+1\right)^2}\,dx$ [Let $u = x^4 + 1$.]

d $\displaystyle\int \frac{1}{\sqrt{4x+3}}\,dx$ [Let $u = 4x + 3$.]

e $\displaystyle\int \sin^2 x \cos x\,dx$ [Let $u = \sin x$.]

f $\displaystyle\int \tan^3 x \sec^2 x\,dx$ [Let $u = \tan x$.]

10 Evaluate each definite integral using the given substitution.

a $\displaystyle\int_{-1}^{0} x^2\left(1+x^3\right)^4\,dx$ [Let $u = 1 + x^3$.]

b $\displaystyle\int_{0}^{\frac{\pi}{2}} \cos^3 x \sin x\,dx$ [Let $u = \cos x$.]

c $\displaystyle\int_{1}^{\sqrt{2}} x\sqrt{x^2-1}\,dx$ [Let $u = x^2 - 1$.]

d $\displaystyle\int_{1}^{e} \frac{(\ln x)^2}{x}\,dx$ [Let $u = \ln x$.]

e $\displaystyle\int_{\frac{1}{2}}^{1} \frac{e^{\frac{1}{x}}}{x^2}\,dx$ [Let $u = \frac{1}{x}$.]

f $\displaystyle\int_{0}^{\frac{\pi}{8}} \frac{\sec^2 2x}{1+\tan 2x}\,dx$ [Let $u = \tan 2x$.]

11 Find each indefinite integral using the given substitution.

a $\displaystyle\int \frac{x}{x-1}\,dx$ [Let $x = u + 1$.]

b $\displaystyle\int \frac{x-1}{\sqrt{x+2}}\,dx$ [Let $x = u - 2$.]

c $\displaystyle\int x\sqrt{2x+1}\,dx$ [Let $x = \frac{1}{2}u^2 - \frac{1}{2}$.]

d $\displaystyle\int \frac{1}{4+\sqrt{x}}\,dx$ [Let $x = (u-4)^2$.]

12 Evaluate, using the given substitution:

a $\displaystyle\int_1^2 x(x-1)^4\,dx$ [Let $x = u+1$.]

b $\displaystyle\int_1^5 \frac{x}{x+3}\,dx$ [Let $x = u-3$.]

c $\displaystyle\int_0^{15} \frac{x}{\sqrt{x+1}}\,dx$ [Let $x = u^2-1$.]

d $\displaystyle\int_2^3 \tfrac{1}{2}x\sqrt{x-2}\,dx$ [Let $x = u^2+2$.]

13 a State the domain and range of the function $y = \sqrt{9-x}$.

b Sketch the graph of the function.

c Calculate the area of the region bounded by the curve and the coordinate axes.

d Calculate the volume of the solid formed when this region is rotated about:
(i) the x-axis, (ii) the y-axis.

14 A horn is created by rotating about the x-axis the region between the curve $y = \dfrac{1}{\sqrt{4-x}}$ and the x-axis from $x = 0$ to $x = 3\frac{3}{4}$. Find the volume of the horn.

15 A vase is designed by rotating the parabola $y^2 = 18(x-6)$ from $y = -6$ to $y = 6$ about the y-axis. Find the exact volume of the vase.

16 a Write $\cos^2 2x$ in terms of $\cos 4x$.

b The region bounded by the curve $y = \cos 2x$ and the x-axis from $x = -\frac{\pi}{6}$ to $x = \frac{\pi}{6}$ is rotated about the x-axis. Show that the solid generated has volume $\dfrac{\pi}{24}\left(4\pi + 3\sqrt{3}\right)\ u^3$.

17 A tank is created by rotating about the x-axis the region between the curve $y = 1 + e^{-x}$ and the x-axis from $x = 1$ to $x = 3$. Find its volume correct to two decimal places.

18 a Sketch $y = 1 - \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and shade the region R bounded by the curve and the coordinate axes.

b Find the volume generated when R is rotated about the x-axis.

19 a Find the stationary points of the curve $y = x + \dfrac{1}{x}$ and sketch its graph.

b Show that the line $y = \frac{5}{2}$ intersects the curve when $x = \frac{1}{2}$ or $x = 2$.

c Find the volume of the solid generated when the region between the curve and the line $y = \frac{5}{2}$ is rotated about the x-axis.

21 The region under the graph $y = 2^{x+1}$ between $x = 1$ and $x = 3$ is rotated about the x-axis.

a Write down the definite integral representing the volume of the solid that is formed.

b Use the trapezoidal rule with five function values to approximate the volume of the solid, giving your answer as multiple of π.

c Find the exact value of the volume, then approximate it as an integer multiple of π. Why is the exact answer smaller than the trapezoidal-rule approximation?